
ARTICLE

Water resources in Spain: addressing challenges to water security

Los recursos de agua en España: abordando los desafíos para la seguridad hídrica

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Abstract: Successfully addressing and resolving water challenges, in a global context where the vectors of population growth, urbanisation, climate change and globalisation are at work, is a key element to foster sustainable development and achieve an improvement in the quality of life of societies and people. It is essential that water is on the political agenda and that many actors play their role properly, with the appropriate governance. The Sustainable Development Goals are not always synergistic among them, and SDG 6 on water interacts with a number of other SDGs. Understanding this is essential to guide new perspectives that adequately combine actions to ensure water security and properly address the ecological services that water provides, supplying the necessary resource for the economic activities that lead to this improvement in the quality of human life. In this regard, the water, food and energy nexus need to be given special consideration. As in other fields, it is necessary to “think global, act local”. And in water, where the processes of resource provision require time to develop substantial actions and high investments, planning processes are the best way to promote consensus and agree on the actions to be carried out. In addition, not just planning, but then implementing these plans diligently and effectively. In Spain, we have been living off our income for years and it is time to move decisively into action, which requires an increase in human and material resources, progress in technology, innovation, and increase in funding.

Keywords: water; sustainable development; water-food-energy nexus; planning; investments; governance

Key points:

Water is a renewable resource needing integrated hydrological management and participatory planning to prevent risks and anticipate measures.

Spain's uneven water flow and shifting demand require greater investment in infrastructure and protective measures than elsewhere in Europe.

Hydrological plans define measures to ensure resource protection and availability; failure to implement them justifies these key investments.

Resumen: El afrontar y resolver con éxito los retos del agua, en un contexto mundial donde actúan los vectores de crecimiento poblacional, urbanización, cambio climático y globalización, resulta un elemento clave para propiciar un desarrollo sostenible y alcanzar una mejora de calidad de vida de las sociedades y las personas. Es imprescindible que el agua esté en la agenda política y que numerosos actores desempeñen adecuadamente su función, con la gobernanza adecuada. Los Objetivos de Desarrollo Sostenible no son siempre sinérgicos entre ellos, y el ODS 6, dedicado al agua, interactúa con un buen número de otros ODS. Comprender esto es esencial

para orientar nuevas perspectivas que combinen adecuadamente las actuaciones para garantizar la seguridad hídrica y atender debidamente a los servicios ecológicos que presta el agua, suministrando el recurso necesario para las actividades económicas que propician esa mejora de la calidad de vida humana. En este sentido, hay que considerar especialmente el nexo entre agua, alimentos y energía. Como en otros campos, es necesario “pensar en global, actuar en local”. En el agua, donde los procesos de provisión del recurso requieren tiempo para desarrollar cuantiosas actuaciones y elevadas inversiones, los procesos de planificación son los indicados para promover consensos y acordar las actuaciones que se van a llevar a cabo para, después, hacer realidad esos planes. En España, llevamos años viviendo de las rentas. Es hora de pasar decididamente a la acción, lo que requiere incrementar los medios humanos y materiales, avanzar en tecnología e innovación, e incrementar la financiación.

Palabras clave: agua; desarrollo sostenible; nexo agua-alimentos-energía; planificación; inversiones; gobernanza.

Puntos clave:

El agua es un recurso renovable que requiere gestión integrada del ciclo hidrológico y planificación participativa para prevenir riesgos y anticipar medidas.

El flujo de agua irregular y la demanda variable en España implican mayor inversión infraestructural y medidas de protección que en el resto de Europa.

Los planes hidrológicos garantizan la protección y disponibilidad del recurso; no aplicar sus medidas justifica inversiones clave añadidas.

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Supplementary information ↓

1. OBJECTIVES, METHODOLOGY AND DATA USED

The main purpose of this article is to draw society's attention to the critical role that water resources play in its present and future and the most important challenges to be resolved, focused in Spain

River basin planning is the main instrument for achieving the good condition of water bodies and meeting water demands through sustainable management of the resource, following the guidelines of Spanish water legislation and European law on this matter. From this perspective, the objective is to analyze the process of preparing hydrological plans in Spain—as an essential tool for defining the goals to be achieved in terms of water resource management—and the results obtained in the different planning cycles.

This will allow for a diagnosis of the effectiveness of the proposed measures and the results obtained, as well as the main challenges to be resolved.

The methodology followed for its development have been very simple, essentially consisting of the following steps:

General review of water resources worldwide and their specific focus on the Spanish case.

Summary of water demands for different uses.

Analysis of resource availability, shortages, or deficits to address the various uses, and environmental limits to be established to ensure sustainable use. Prospects and impacts of climate change. Conflicts between water uses, sustainability, and resource preservation.

Review of the process for drafting hydrological plans in Spain—in accordance with EU water legislation—as a tool for diagnosing problems and developing the corresponding Programs of Measures to achieve established environmental objectives.

A brief analysis of the results achieved with the first and second cycle plans and the trends observed.

An analysis of the main contents of the plans currently in force. A brief analysis of the results and the future uncertainties to be clarified.

A reasoned diagnosis of the main challenges to be resolved in water resource management for sustainable use and to ensure adequate availability, as well as criteria and proposals for overcoming these challenges. Special attention is paid to the investments to be made, justifying their need as well as the expected benefits of said investments.

And, finally, a section of conclusions.

The breadth and diversity of the information involved in this analysis means that the documentation and data used must come from a variety of sources, preferably using their original sources (official networks of the institutions responsible for their registration) and recognized reliable organizations, when these are not directly available.

2. INTRODUCTION AND GLOBAL CONTEXT

Water is critical to all environmental and social systems that underpin the global economy. Therefore, the critical importance of water must be recognised in line with the economic development it brings, as well as its social and environmental benefits.

In today's world we can observe vectors that have a major impact on water and water security:

- Population growth and urbanisation: Cities and neighbourhoods are expanding rapidly, and this phenomenon has different characteristics from the past urbanisation process: in many cities, urbanisation is uncontrolled and without an economic basis. Therefore, preparations are needed to ensure the planning and provision of basic services in cities.

Population growth, land use change, demographic change, including migration and urbanisation are among several social stressors that exacerbate water resources vulnerability and have a concomitant impact on water access and availability. Without effective institutions and adaptive management practices, vulnerable countries will continue to experience adverse impacts from climate change. An enabling environment for effective adaptation through multiple levels of intervention implies responsive, accessible, inclusive and equitable institutional capacity, especially in the context of governance, capable of promoting change at local, provincial, national and regional levels and beyond. Transboundary cooperation is also crucial to limit the vulnerability of water resources.

- Climate change: Although much of the scientific community agrees on climate change, the exact impact of these changes on the hydrological cycle is not well predictable. Water managers must learn to deal with uncertainties. Climate change calls for flexibility in response efforts.

The European Union and the World Meteorological Organisation (WMO) act through various programmes at local and river basin level for the implementation of adaptation measures, capacity building and the development of the climate information service.

Climate change adaptation strategies need to be developed: water cuts across most adaptation policies and needs to become a major focus of the Climate Convention's concerns.

Impacts on regional water accessibility and availability could lead to regional water crises, resulting in destabilisation, violence and conflict, which would affect poor and vulnerable people the most.

This has been repeatedly stated in successive IPCC reports and is reflected in its latest publications. (IPCC, 2023: Summary for Policymakers. In: Climate Change 2023: Synthesis Report. Contribution of Working Groups I, II and III to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change [Core Writing Team, H. Lee and J. Romero (eds.)]. IPCC, Geneva, Switzerland)

- Globalisation and improvement of the human development index HDI: The process of globalisation involves large exchanges of virtual water based on international trade in food, goods and services, as well as a process of increasing demands for water to meet new needs and thus improve the quality of life in developing and underdeveloped countries.

Globalisation also extends to the financial markets, and in this sense, it should be stressed that for the international community and investors to allocate economic resources to the water sector, it is essential to guarantee legal certainty and to have adequate governance mechanisms that offer the right conditions to guarantee the success of the actions and, where appropriate, the return of committed funds.

The possible futures in the context of global change are highly complex, as is already evident in a number of current actions.

Part of the solution is to ensure that water is brought to its rightful place on the political agenda and on the climate change agenda, in order to get proper recognition. Water is a cross-cutting resource, not a separate sector, and water expertise needs to be represented where decisions are made. Furthermore, water issues need to be addressed through the establishment of local, regional and global levels.

In this regard, the consideration of SDG 6 “Safe water and sanitation”, which has made it possible to identify the main challenges at the global level, monitor them and propose initiatives that respond to them in the 2030 Agenda, is to be welcomed:

- 85% of the population lives in arid areas and by 2030, half of the population will live in areas of high-water stress.
- 2.2 billion people do not have access to safe and adequate water
- More than 800 million people suffer from malnutrition.
- 4.2 billion people do not have access to adequate hygiene and sanitation services
- Between 6 and 8 million people die each year from water-related disasters.
- 85% of wastewater is discharged without adequate pre-treatment
- In 30 years, there has been a 50% decline in freshwater animal species.
- There are 600 transboundary aquifers, shared between 2-4 states.

Climate change and the other vectors of this global change pose a challenge to water managers and users, as well as policy makers at different levels. Given the intrinsic link between other sectors and freshwater resources and ecosystems, increased vulnerability of water resources due to climate change may affect, inter alia: ecosystems and biodiversity; agriculture and food security; land use and forestry; drinking water and sanitation; health; urban settlements and infrastructure; and power generation and energy supply.

Strategies to respond to this global change can include different components: investment in infrastructure, integrated water resources planning and management (IWRM), risk assessment and analysis; improved methods, management and decision-making capacity; and participation in decision-making by policy makers, practitioners, researchers and vulnerable communities, among other stakeholders. Building resilience involves a combination of bottom-up and top-down approaches, early warning systems and disaster risk management strategies, which offer short- and medium-term development benefits and reduce long-term vulnerability. Figure 1 shows some of the main problems caused by this global change and a guide to possible solutions to address them.

The development of scenarios and models should be driven through stakeholder participation to ensure relevance and pertinence to decision-makers. These scenarios and models need to be done on a temporal and geographical scale for the different disciplines in order to address particular challenges.

Furthermore, as shown graphically in Figure 2, there are other SDGs closely related to water, which do not necessarily have positive synergies with SDG 6, requiring a proper balance between the corresponding sectoral policies, which is not necessarily easy and requires good governance, good governance and decision-making capacity.

In this regard, it is particularly necessary to consider the water-food-energy nexus (Fig. 3). Water is a necessary resource to provide irrigation for agricultural and livestock production and to achieve food security. It is the most water-demanding sector in the world. And as for energy, which is essential for industrialisation, economic growth and improving the quality of life, in addition to hydroelectric production and the cooling of thermal power plants, water through pumped hydroelectric power plants must provide the necessary storage for other renewable, non-manageable energies (photovoltaic and wind) in order to make decisive progress in decarbonising the economy and mitigating climate change). In addition, water consumption will grow for the production of green hydrogen or for cooling data centres, which are experiencing major growth as a result of the digitalisation process and artificial intelligence mechanisms.

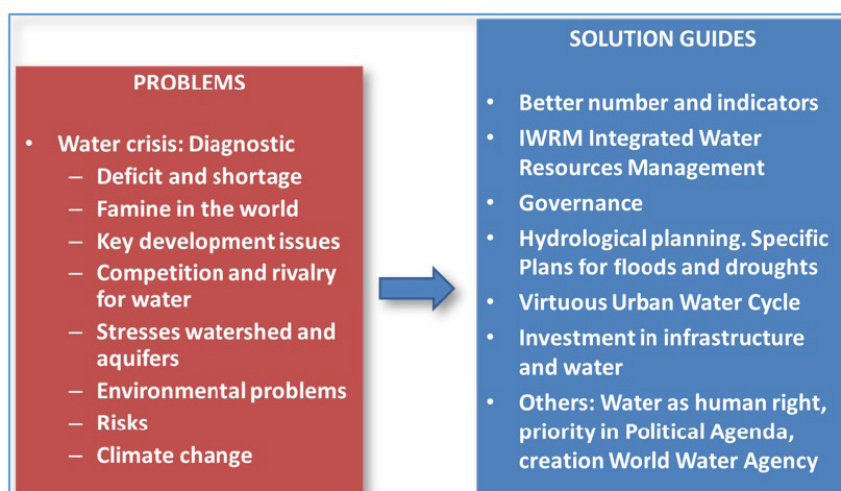


Figure 1. Challenges and guide to solutions to water problems.

Figura 1. Retos y guía de soluciones a los problemas del agua.

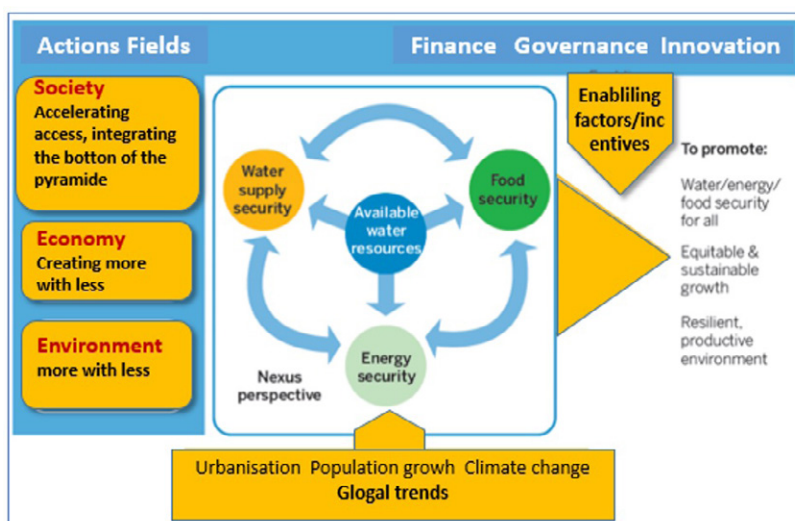


Figure 3. Water-Food-Energy Nexus. Source: H. Hoff (2011) (Stockholm Environment Institute, SEI).

Figura 3. Water-Food-Energy Nexus. Fuente: H. Hoff (2011) (Stockholm Environment Institute, SEI).

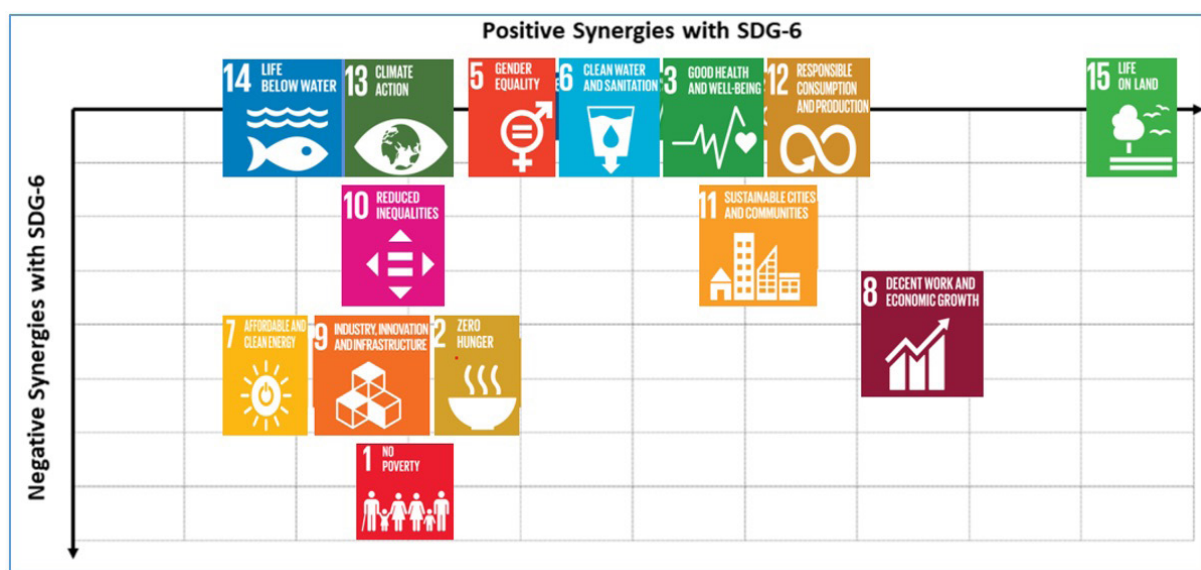


Figure 2. Positive and negative synergies (conflicts) of the SDGs with SDG-6. The horizontal axis shows positive synergies (synergies increase from left to right) and the vertical axis shows negative synergies/conflicts (synergies increase from top to bottom). The graph is based on information from UN-Water (UN-Water, 2016 Water and Jobs). WFEU Water WG Report (2019) World Federation of Engineering Organizations (WFEU) Working Group on Water (WGoW).

Figura 2. Sinergias positivas y negativas (conflictos) de los ODSs con el ODS-6. En el eje horizontal se expresan las sinergias positivas (las sinergias aumentan de izquierda a derecha) y en el vertical las negativas/conflictos (las sinergias se incrementan de arriba abajo). El gráfico se ha elaborado en base a la información de UN-Water (UN-Water, 2016 Water and Jobs). I WFEU Water WG Report (2019) World Federation of Engineering Organizations (WFEU) Working Group on Water (WGoW).

3. SPAIN AND WATER

3.1. Resources and uses

Spain is marked by great temporal and geographical irregularity in terms of rainfall and the availability of water resources. This makes it unique in the framework of Europe in general and the European Union in particular, where concern is focused on water quality (and not quantity), as can be seen in the Water Framework Directive of 2000, and the Directives related to sanitation and wastewater treatment (Directive 91/271/EEC on urban wastewater treatment, currently in the process of renewal) and subsequently to protection against the risk of flooding, following the major floods in central Europe in 2005.

Indeed, while in Europe the main figures (Fig. 4) show that only 40% of water abstractions (of the order of 243,000 hm³ /year) are used for agriculture, and only 10% of the water used is regulated in reservoirs, in Spain, water withdrawals are in the order of 30,000 hm³ /year, almost 80% is used for irrigation, and around 75% of the water used is thanks to the regulation offered by reservoirs.

It is enough to compare the correlation with the population to understand the importance and scope of this geographical condition and what it means in terms of additional effort for our country's competitiveness and the quality of life of our inhabitants: With 9% of the population, Spain extracts 12% of the resources of the EU total. This is mainly due to the fact that in the rest of the EU there is a much higher percentage of "green water" (rainwater) in the water used for crops, and in Spain there is a higher percentage of "blue water" (which requires regulation, transport and distribution). Furthermore, the level of resource use is higher in Spain than in the rest of the countries, including those in the European Mediterranean region (Table 1).

In order to be able to use water for the socio-economic development of our country, it has been essential to carry out major hydraulic works and develop a powerful system of governance. These works are still essential today. These hydraulic works, in particular, and the Spanish Water Governance System in which they are integrated, in general, have made it possible to meet, with average natural resources of about 110 km³, a demand of approximately 30 km³, multiplying the availability of water for different uses by 4.

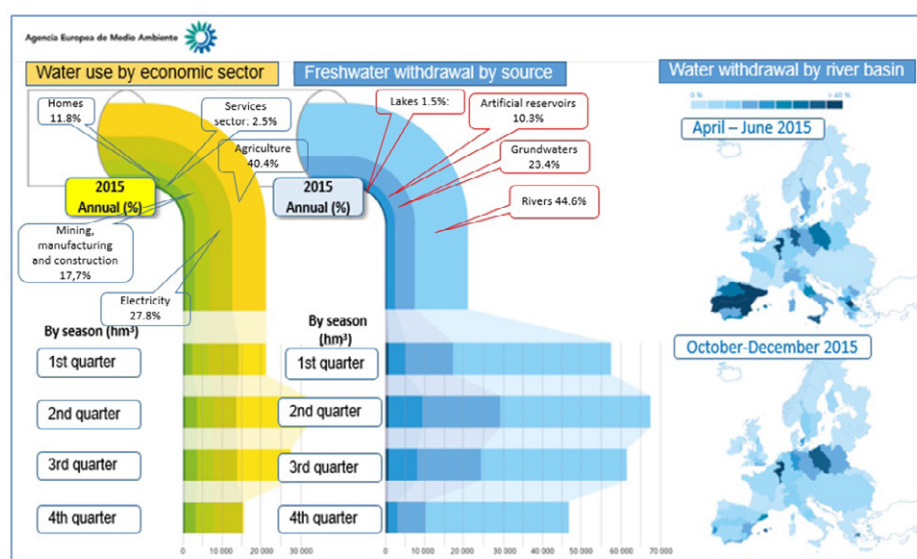


Figure 4. Source: European Environment Agency (EEA, 2018), “Water use in Europe - Quantity and quality face big challenges”.

Figura 4. Fuente: Agencia Europea de Medioambiente, (EEA, 2018), “Water use in Europe - Quantity and quality face big challenges”.

Table 1. Source: Centro de Estudios y Experimentación CEDEX (2000).

Tabla 1. Fuente: Centro de Estudios y Experimentación CEDEX (2000).

COUNTRY	INTERNAL/ POTENTIAL RESOURCES (hm ³ /year)	TOTAL/ POTENTIAL DEMAND (hm ³ /year)	EXPLOITATION INDEX	CONSUMPTION INDEX
SPAIN	111.186 / 88.949	35.323 / 20.784	0,40	0,23
FRANCE	185.000 / 148.000	39.657 / 7.044	0,27	0,05
ITALY	155.000 / 124.000	43.694 / 19.375	0,35	0,16
GREECE	52.550 / 42.040	7.285 / 5.212	0,17	0,12

According to the figures shown in Figure 5, the situation in northwest Spain (many resources, low demand for blue water) has nothing to do with the situation in the southeast (very few resources, significantly higher demand), even in average values.

In our country, 30.74% of the average natural resource is allocated to meet gross water demands for consumptive uses; 24.88% for agriculture; 4.74% for urban supply; and 1.12% for industries not included in urban supply and miscellaneous uses. The most important consumptive use of water is agricultural, well ahead of urban supply and industrial use (Fig. 6).

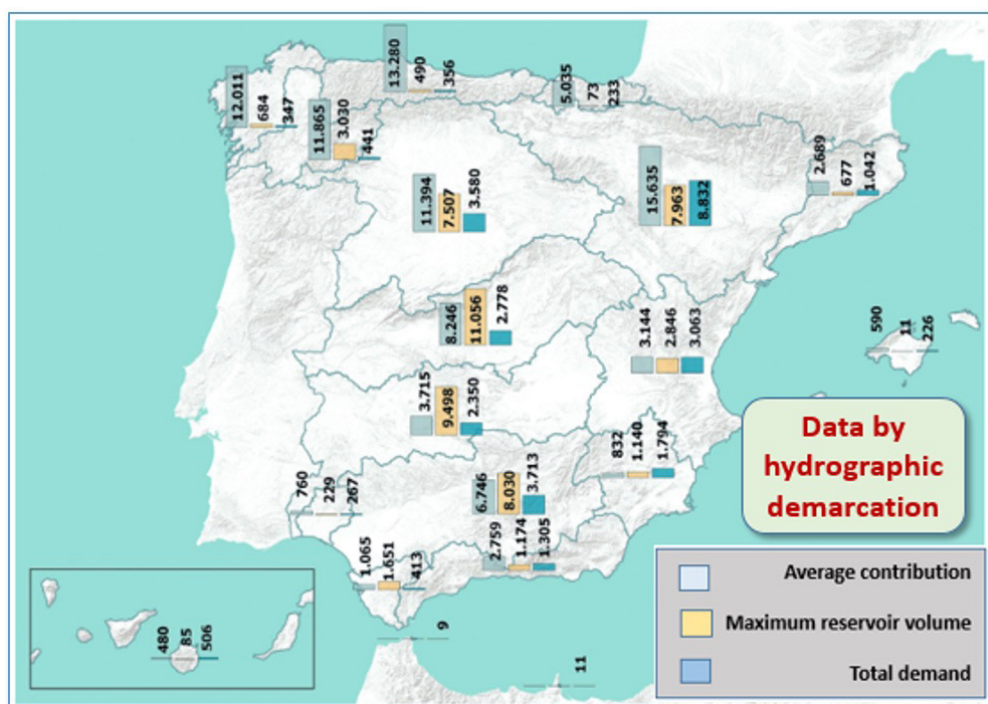


Figure 5. Ministerio para la Transición Ecológica y el Reto Demográfico MITECO. Dirección General del Agua (2022). Annual report on water use in Spain 2021-22.

Figura 5. Ministerio para la Transición Ecológica y el Reto Demográfico MITECO. Dirección General del Agua (2022). MITECO, Informe anual de los usos del agua en España 2021-22.

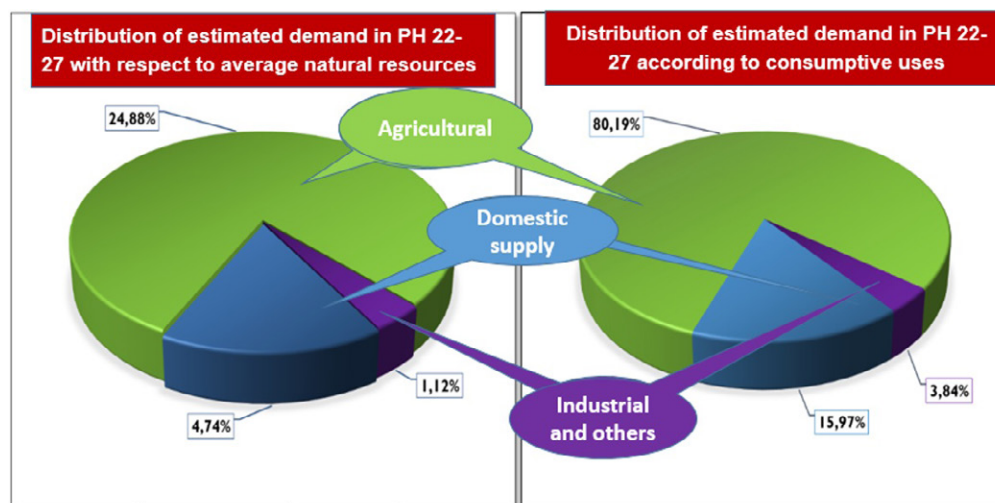


Figure 6. Consumptive use of water in Spain. Ministerio para la Transición Ecológica y el Reto Demográfico MITECO. Dirección General del Agua (2022). Consumer use. Annual report on water use in Spain 2021-22.

Figura 6. Ministerio para la Transición Ecológica y el Reto Demográfico MITECO. Dirección General del Agua (2022). Usos de consumo del agua en España. Informe anual de los usos del agua en España 2021-22.

Although we must not forget the importance of other non-consumptive uses, such as energy use in hydroelectric power station.

With regard to their typification in terms of what they use the water for, each river basin has its own vocation (Fig. 7). Thus, the Cantabrian coast, the internal basins of Catalonia and the island territories (Balearic and Canary Islands) emphasise urban and industrial use, while in the large inland basins and those of the southeast, agricultural use predominates.

On the other hand, there are also notable differences in terms of the origin of the resource (Fig. 8). More than 72% (22,207 hm³) comes from rivers, regulated by more than 1,110 large reservoirs (with a capacity of 61,300 hm³), while 21% (6,547 hm³) comes from groundwater. Although in comparative figures it is much smaller, the role played by non-conventional resources (398 hm³ from reuse and 658 hm³ from desalination) and inter-basin transfers (805 hm³) is fundamental for water security, and is set to increase in the future.

It should be pointed out that Spain is a “reservoir-dependent” country and, as the studies included in the White Paper on Water (Ministry of the Environment, 2000) showed, without them less than 8 km³ (25% of current use) could be supplied with adequate guarantees, and it would not be able to meet the quality of life conditions for the current population. In more current studies (Special Drought Plans) this dependence on the regulation of reservoirs is again made clear, since more than 90% of the scarcity indicators that are collected for the different exploitation systems are the volume stored in the reservoirs of these systems and, to a lesser extent, the piezometric levels of the groundwater bodies, as well as other composite indices.

However, it is worth highlighting the important role that groundwater resources play in ensuring a safe and quality water supply. The ubiquity of groundwater, which obviates the need for large pipelines to transport water from the source of the resource to the point of use, the large volume of water reserves stored in aquifers, and the natural purification capacity developed through the infiltration-recharge process, give groundwater a unique interest and strategic value for its use, with significant advantages in terms of efficiency and environmental impact.

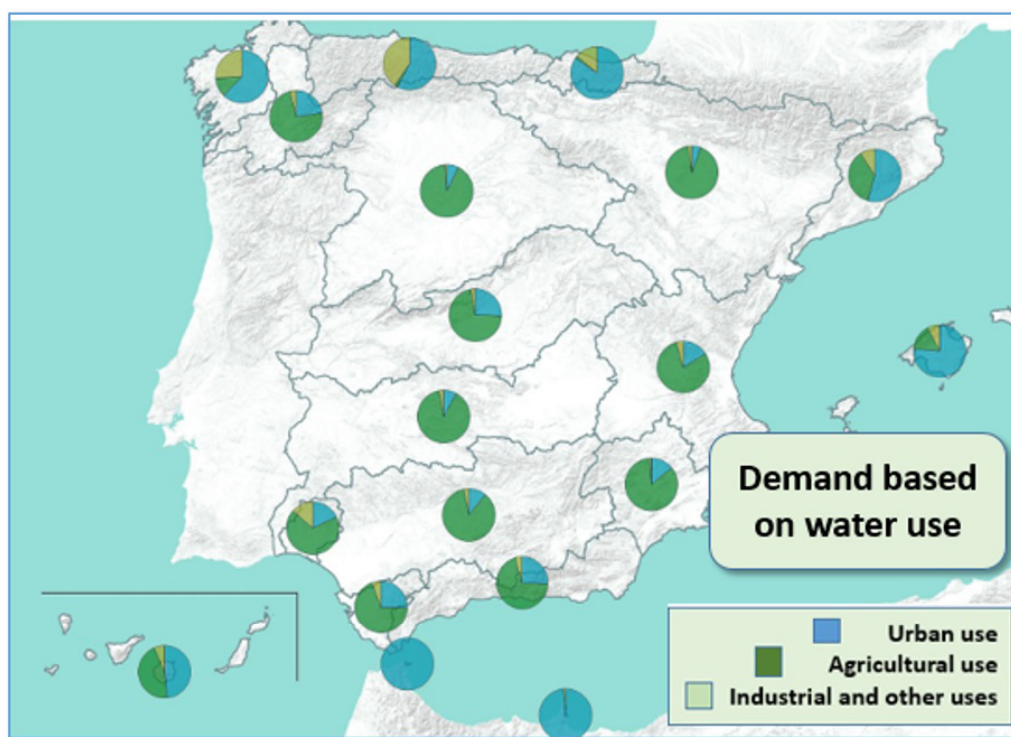


Figure 7. Distribution of consumptive demands according to water uses and their distribution by River Basin Districts. Ministerio para la Transición Ecológica y el Reto Demográfico MITECO Annual report on water use in Spain 2021-22.

Figura 7. Reparto de demandas consuntivas según usos del agua y su distribución por Demarcaciones Hidrográficas. Ministerio para la Transición Ecológica y el Reto Demográfico MITECO Informe anual de los usos del agua en España 2021-22.

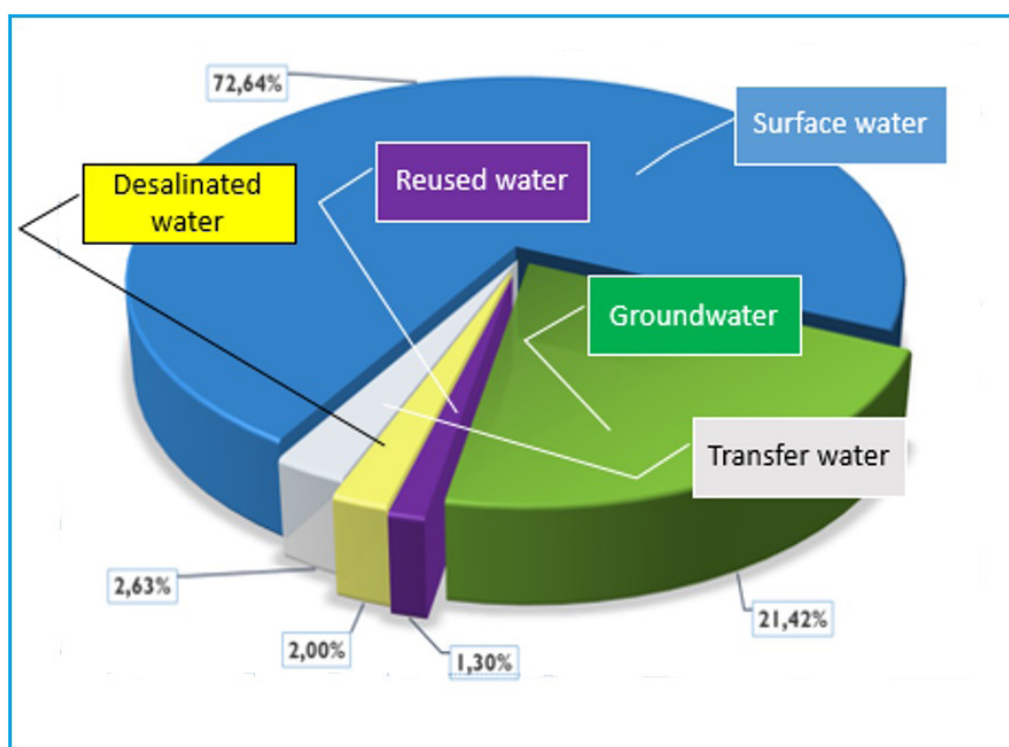


Figure 8. Percentage distribution of water demand by origin of water resources. Ministerio para la Transición Ecológica y el Reto Demográfico MITECO, Annual report on water use in Spain 2021-22.

Figura 8. Distribución porcentual de las demandas de agua según origen del recurso hídrico. Ministerio para la Transición Ecológica y el Reto Demográfico MITECO Informe anual de los usos del agua en España 2021-22.

3.2. Water quality

One of the main challenges of water management in Spain is to achieve good status of water bodies (rivers, lakes, lagoons, aquifers...), in accordance with the provisions of Spanish legislation, in application of the WFD.

For inland waters, this objective implies that surface water bodies identified in the hydrological plans should achieve good chemical status and good ecological status or, where appropriate, good potential status in the case of heavily modified or artificial water bodies. For groundwater bodies, good chemical status and good quantitative status must be achieved so that the long-term average annual rate of abstraction does not exceed that of the available water resources and does not lead to anthropogenic alterations that would prevent the achievement of the environmental objectives for the associated surface waters.

The following graphs (Figures 9, 10, 11 y 12) show the initial and projected situation at the end of the previous planning cycle (second cycle 2017- 2021), as well as the results of the annual monitoring reports. These show a stagnation, if not a setback, in the evolution of water quality and the gap to reach the proposed targets by 2021.

Groundwater presents its own problems, particularly when the good quantitative status of a groundwater body is not reached, associated with the risks of overexploitation (which in coastal areas favours saline intrusion, with consequences that are difficult to reverse). Or diffuse pollution, when the good chemical status is endangered, specially that induced by the overuse of fertilisers in certain agricultural practices, which leads to a large number of groundwater bodies exceeding the permitted limits for nitrate content and to a proliferation of declarations of vulnerable zones.

The solution to this problem represents an enormous effort that still needs to be made to achieve good (overall) status of all water bodies (surface and groundwater), and the little effective progress that has been made through the various planning cycles.

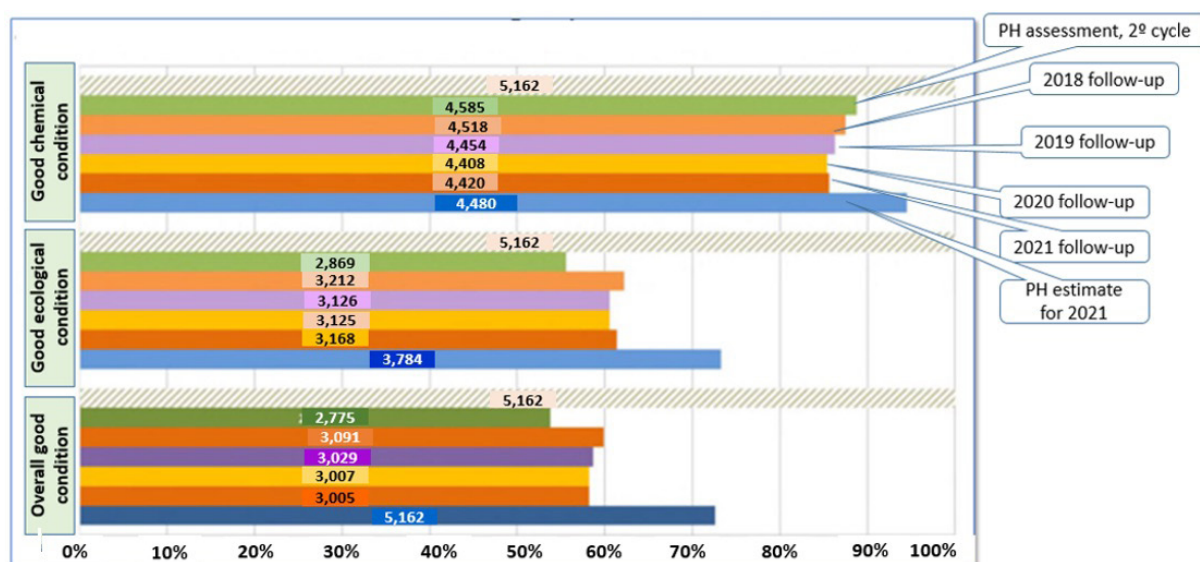


Figure 9. Water quality: surface water bodies. Comparison of surface water bodies in good chemical, ecological and overall status with respect to the total and for different scenarios: time of preparation of the second cycle plans, indicative estimate in the years 2028, 2029, 2020 and 2021, and targets for 2021 foreseen in the second cycle plans. (MITECO Síntesis de los planes hidrológicos españoles. Segundo Ciclo de la DMA (2015-2021 and own elaboration).

Figura 9. Calidad del agua: Masas de agua superficiales. Comparativa de masas de agua superficial en buen estado químico, ecológico y global con respecto al total y para distintos escenarios: momento de elaboración de los planes del segundo ciclo, estimación orientativa en los años 2028, 2029, 2020 y 2021, y objetivos para 2021 previsto en los planes del segundo ciclo. (MITECO Síntesis de los planes hidrológicos españoles. Segundo Ciclo de la DMA (2015-2021 y elaboración propia).

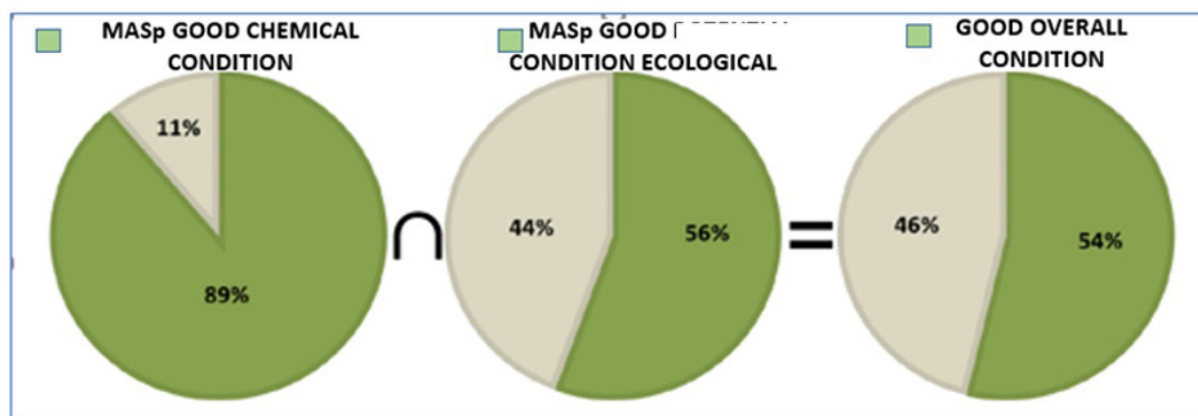


Figure 10. Surface water bodies of good chemical status, good ecological status or potential and good (overall) body status in second cycle plans. (Ministerio para la Transición Ecológica y el Reto Demográfico MITECO Síntesis de los planes hidrológicos españoles. Segundo Ciclo de la DMA (2015-2021 and own elaboration).

Figura 10. Masas de agua superficial en buen estado químico, buen estado o potencial ecológico y buen estado de la masa (global) en los planes del segundo ciclo. (Ministerio para la Transición Ecológica y el Reto Demográfico MITECO Síntesis de los planes hidrológicos españoles. Segundo Ciclo de la DMA (2015-2021 y elaboración propia).

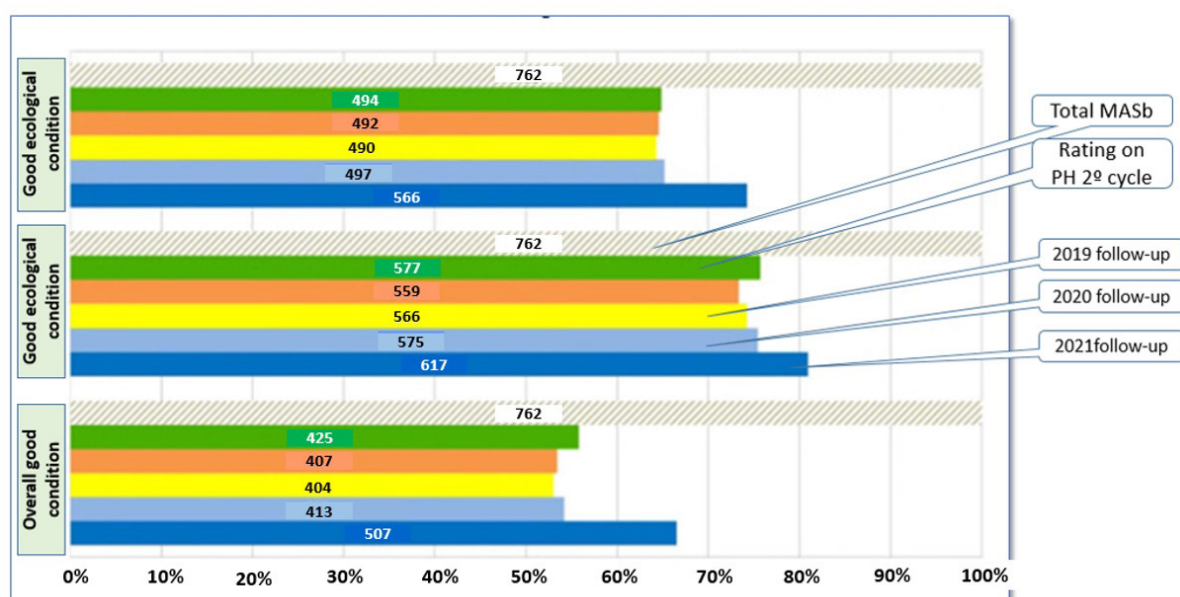


Figure 11. Water quality: groundwater bodies. Comparison of groundwater bodies in good chemical, quantitative and global status with respect to the total and for different scenarios: time of preparation of the second cycle plans, indicative estimation in the years 2028, 2029, 2020 and 2021, and Targets for 2021 foreseen in the second cycle plans. (Ministerio para la Transición Ecológica y el Reto Demográfico MITECO Síntesis de los planes hidrológicos españoles. Segundo Ciclo de la DMA (2015-2021 and own elaboration).

Figura 11. Calidad del agua: agua subterránea. Comparativa de masas de aguas subterráneas en buen estado químico, cuantitativo y global con respecto al total y para distintos escenarios: momento de elaboración de los planes del segundo ciclo, estimación orientativa en los años 2028, 2029, 2020 y 2021, y Objetivos para 2021 previsto en los planes del segundo ciclo. (Ministerio para la Transición Ecológica y el Reto Demográfico MITECO Síntesis de los planes hidrológicos españoles. Segundo Ciclo de la DMA 2015-2021 y elaboración propia).

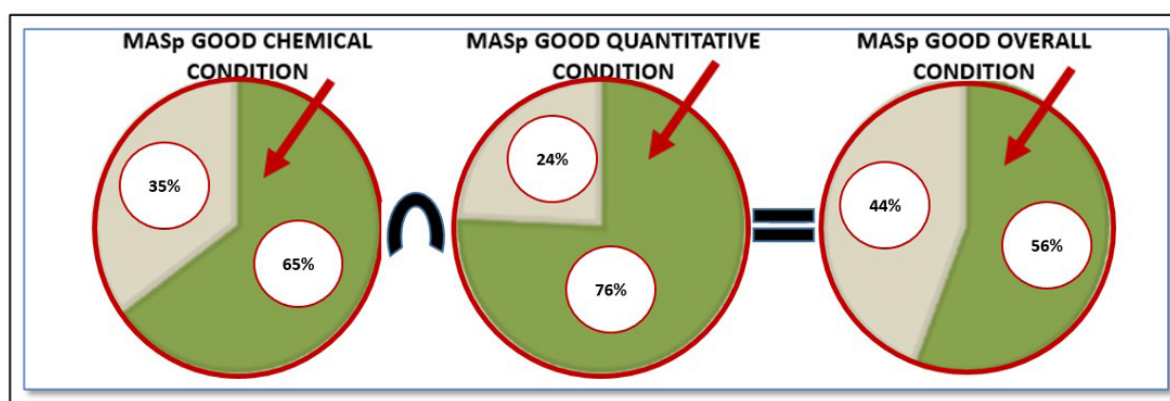


Figure 12. Groundwater bodies of good chemical status, good quantitative status and good (overall) body status in second cycle plan (Ministerio para la Transición Ecológica y el Reto Demográfico MITECO Síntesis de los planes hidrológicos españoles. Segundo Ciclo de la DMA (2015-2021 and own elaboration).

Figura 12. Masas de agua subterránea en buen estado químico, buen estado cuantitativo y buen estado de la masa (global) en los planes del segundo ciclo (Ministerio para la Transición Ecológica y el Reto Demográfico MITECO Síntesis de los planes hidrológicos españoles. Segundo Ciclo de la DMA (2015-2021 y elaboración propia).

This will require better control of discharges into the public water and maritime-terrestrial domain (identification of unauthorised discharges, monitoring of authorised discharges, sanctioning regime...) and a careful analysis of the emission limit values in the discharge authorisations, when it is necessary

to establish more demanding limits in order to achieve the good chemical status of the water body receiving the discharge (appropriate emission limit), in accordance with the provisions of the Environmental Quality Standards (EQS). And, therefore, more intensive treatment.

Of a total of 20,508 hm³ (Fig. 13) corresponding to authorised water discharges, 3,841 hm³ are of urban origin (19%) and 16,667 hm³ are of an industrial nature (81%) which discharge directly (after treatment) into water bodies. 63% of these effluents discharge into surface waters and 3% into groundwater. The remaining 34% corresponds to discharges into coastal or transitional waters.

Some 68% of urban discharges originate from domestic uses and 14% are of an industrial type, due to activities of this nature carried out within the urban fabric that are discharged through the sewage network, in the conditions required for this purpose. The remaining 17% corresponds to municipal activities and services, such as street cleaning, garden watering, etc.

These percentages of urban and industrial discharges, aggregated at the national level, have a very diverse distribution when analysed at the scale of River Basin Districts (as shown in Figure 13), revealing the different productive structure of their territories and the use of water derived from it.

Figure 14 shows the results for the different types of industrial discharges, classified by the type of industry they originate from, aggregated at the national level. This allows for an analysis of the importance of each industry category in the total amount of wastewater discharged, after treatment.

It is estimated (iAgua, 2018; <https://www.iagua.es/blogs/pablo-gonzalez-cebrian/cual-es-antiguedad-red-alcantarillado-agua-espana>) that, in Spain, the length of urban sewerage networks reaches 165,000 km. The annual volume of treated urban wastewater exceeds 4,100 hm³, treated in more than 2,000 Wastewater Treatment Plants (WWTP),

The European Commission has opened several investigation and sanction proceedings against Spain, mainly in relation to the degree of compliance with Community legislation on urban waste water treatment (Directive 91/271/EEC) and the protection of water against pollution by nitrates from agricultural activities (Directive 91/676/EEC), issues on which the actions undertaken have not been sufficiently effective to obtain the required results and which will have to be addressed in successive hydrological plans.

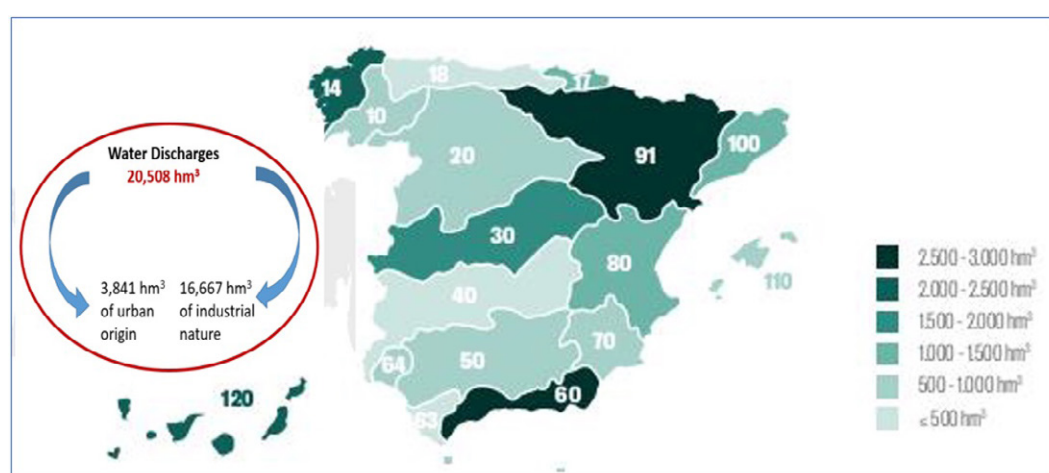


Figure 13. Percentage of urban and industrial discharges. Distribution of total discharge volumes, by River Basin District. Source: Censo Nacional de Vertidos. Ministerio para la Transición Ecológica y el Reto Demográfico (MITECO) (2022) Figure taken through <https://www.iagua.es/noticias/redaccion-iagua/miteco-publica-censo-nacional-vertidos-al-dominio-publico-hidraulico>.

Figura 13. Porcentaje de vertidos de naturaleza urbana e industrial. Distribución de los volúmenes totales de vertido, por Demarcaciones Hidrográficas. Fuente: Censo Nacional de Vertidos. Ministerio para la Transición Ecológica y el Reto Demográfico (MITECO) (2022). Figura tomada a través de <https://www.iagua.es/noticias/redaccion-iagua/miteco-publica-censo-nacional-vertidos-al-dominio-publico-hidraulico>.

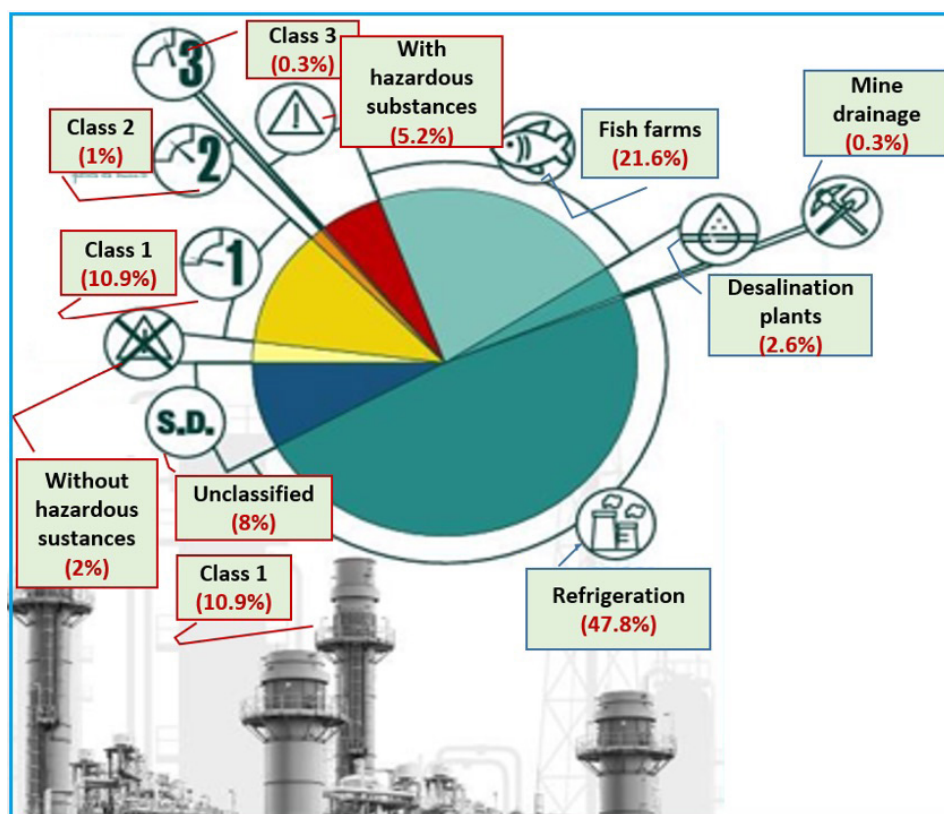


Figure 14. Distribution of the volume of industrial wastewater by type of industry.
Source: Water use reports (2021 - 2022) Spanish Water Governance System (SEGA).

Figura 14. Distribución del volumen de aguas residuales de carácter industrial, por clase de industria.
Fuente: Informes del uso del agua (2021 2022) Sistema Español de Gobernanza del Agua (SEGA).

The gap to achieve good status of water bodies is a challenge at EU level. To address this, the new Waste Water Treatment Directive, (Directive (EU) 2024/3019 of the European Parliament and of the Council of 27 November 2024 on urban wastewater treatment /recast), will increase the requirements in these areas:

- Reduction of the threshold for the implementation of secondary treatment from 2,000 to 1,000 population equivalent (p.e.) in inland waters and from 10,000 to 2,000 p.e. in coastal waters.
- Tertiary treatments for N and P reduction will be implemented for all urban agglomerations UUAA above 150 K p.e and those UUAA above 10 K p.e. that could be at risk will be analysed, which means that tertiary treatments will also be required for many of them.
- Quaternary treatments (4th) will be implemented in UUAA of more than 150 K p.e. (84 UUAA are affected) to reduce the concentration of certain organic micropollutants included in Annex V of the new Directive.
- Integrated water management plans (IWMPs) shall be established to reduce the impact of stormwater discharges. These plans are compulsory for UUAA of more than 100 K p.e., although probable risk plans must be drawn up for UUAA of more than 10 K p.e. In this section it should be borne in mind that the modification of the Public Hydraulic Domain Regulation (RDPH) establishes this limit in UUAA of more than 50 K p.e.
- All Waste Water Treatment Plans (WWTPs) larger than 10 K p.e. will have to implement systems to achieve energy neutrality, starting however with those larger than 100 K p.e. In other words, WWTPs will have to make maximum use of the biogas generated in their own facilities to produce the energy consumed in the treatment process. This objective will imply a major revision of the sludge treatment lines, especially in WWTPs of more than 50 K h-e, many of which do not have

anaerobic digestion systems and not all of which have a gas line to feed energy generation engines. The data available to us suggest that around 60% of the sludge is subjected to this digestion process and a much smaller percentage has energy recovery (so far only available in large plants).

This will mean that there will be an urgent need to introduce new actions and increase the level of investment in water quality in the coming years.

4. RIVER BASIN MANAGEMENT PLANS (PLANES HIDROLÓGICOS) AND PROGRAMS OF MEASURES

4.1. The water planning process in Spain

The management of water resources requires adequate foresight in the adoption of organisational and regulatory measures and the forecasting of actions in the field of hydraulic infrastructures, which will enable future water use needs to be met with guarantees and their adequate protection. This is the only way to ensure sustainable use of the resource.

Water Law 29/1985 consolidated - among other aspects - the principle of organisation and action by river basin, incorporating respect for the integrity of the hydrological cycle and legally regulating the planning of the use and preservation of water resources. This established a model that has served as inspiration in the international context and, in particular, in European legislation.

As a result, four planning cycles have been carried out in Spain: the first corresponding to the River Basin Management Plans (called as Planes Hidrológicos) approved in 1998, in accordance with the legislation in force at the time; and three consecutive planning cycles 2010-2015 / 2016-2021 / 2022-2027 (current), corresponding to six-year periods. These plans cover both inland waters - surface and groundwater - of the river basins, as well as to coastal and transitional waters, constituting, as a whole, the River Basin Districts (Demarcaciones Hidrográficas).

In accordance with the current Water Law, the general objectives of the hidrological plans are to achieve the good state and adequate protection of the public water domain (including the waters integrated in it), the satisfaction of water demands, the balance and harmonisation of regional and sectorial development, increasing the availability of the resource, protecting its quality, economising its use and rationalising its uses in harmony with the environment and the other natural resources.

These plans define the programmes of measures to be carried out in each River Basin District (Demarcación Hidrográfica) to achieve compliance with the environmental objectives (status of the water bodies); and for the protection and sustainable use of the resource, derived from the country's general socio-economic objectives.

4.2. Evolution and progress in planning cycles

In terms of monitoring the degree of implementation of previous plans and taking a general retrospective look at the planning process, there has been a systematic failure to meet the objectives established in previous plans, with many of the measures envisaged not being carried out for various reasons. Fundamentally for budgetary reasons, since the economic forecasts of the plans are not binding and their execution corresponds to different administrations, in addition to the General State Administration (GSA).

Of the total investment foreseen for existing measures (2033) in the second cycle plans, only 8.6% had been completed by the end of the cycle and 14.5% was under implementation.

Considering the total planned investment in existing measures, the budget implemented at the end of 2020 was 13.00%, with 4.3% for completed measures, 1.4% for regular completed measures (maintenance and operation of infrastructures and networks, regular reporting, etc.) and 7.3% for ongoing measures.

The investment executed from 2015 to 2021 and planned to be realised until 2033 is as follow (**Table 2**):

Table 2. Investment executed from 2015 to 2021.

Tabla 2. Inversiones ejecutadas desde 2015 hasta 2021.

Investment (million euros)	
Investment implemented as of December 2020	5,617.24
Completed measures	1,862.53
Completed-periodic	621.33
Measures in progress	3,133.38
Investment to be implemented until 2021	16,024.04
Investment to be implemented until 2023	37,572.53

The deviation of the investment path with respect to that programmed by the plans is particularly affecting, as we have seen, the needs for wastewater treatment and its consequences on the state of the bodies of water. The reasons for the deficient results obtained would merit a detailed analysis, but there is no doubt that the lack of budgetary linking of the plans, the complex distribution of responsibilities for investment (which not only affect the General State Administration) and the difficulty of finding sanctioning or corrective mechanisms for non-compliance by the agents involved, constitute some of the most relevant motivations.

The effect of the actions, which is shown in the medium and long term, works against allocating budgets to these actions, so that whoever launches and starts them does not inaugurate them. In addition, there is the refusal to face the unpopularity that raising water tariffs entails. Tariffs that are in the order of a third of what we all pay for the use of our mobile phones. In this way, a great deal of educational work is needed on the need to invest in water, so that the population is aware of and values resolving these gaps we have to guarantee sustainable development in such a relevant and delicate matter.

4.3. Third Planning Cycle: River Basin Management Plans of River Basin Districts 2022-2027 (Planes Hidrológicos de las Demarcaciones Hidrográficas 2022-2027)

These plans are currently in force. All details regarding their development, processing, public consultation, approval, and content are accessible through https://www.miteco.gob.es/es/agua/temas/planificacion-hidrologica/planificacion-hidrologica/pphh_tercer_ciclo.html

The general structure of these plans comprises an essentially documentary part (Report and Annexes), and a normative part promulgated by Royal Decrees.

In the first one, the data and studies necessary to know the current situation of the River Basin Districts (Demarcaciones Hidrográficas), the degree of compliance with the objectives established in the previous plans that will allow the formulation of the Programme with the relevant measures to achieve the environmental objectives assigned to the bodies of water (and other objectives for the protection and sustainable use of water resources demanded by human needs and the productive sectors, in a justified manner) are collected.

4.3.1. Third planning cycle (2022-2027) Water resources, demands and uses

From the point of view of the assessment of the renewable water resources of each river basin, the new plans confirm the trend towards decreasing values, as shown by the comparison between the average annual values of series of similar length (approximately 40 years) but corresponding to different historical periods, with the most recent showing significantly lower results. This would be consistent with the expected effects of the Climate Change in process. Observed in all basins (Fig. 15).

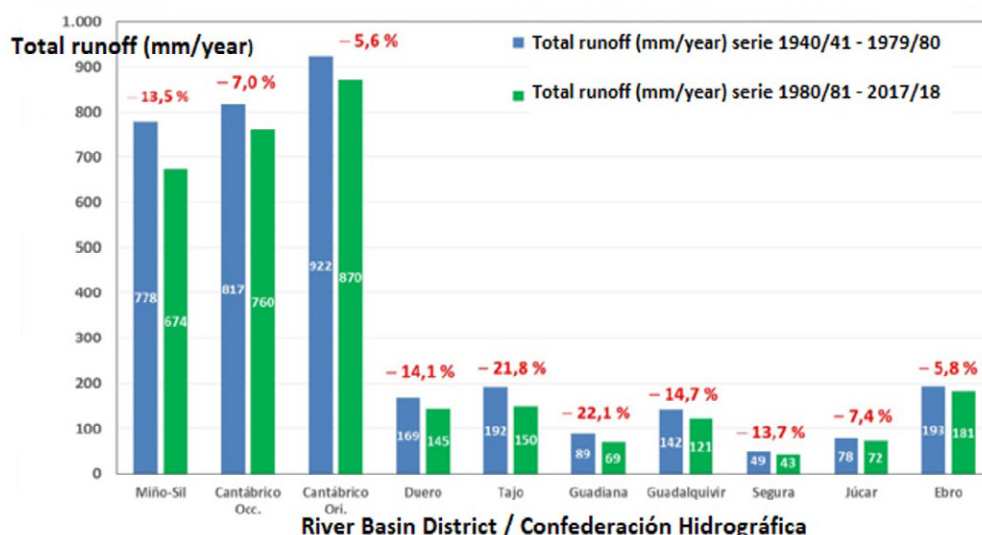


Figure 15. Comparison among River Basin District runoffs, 1940/41 series, 1979/80 series and 1980/81 series - 2017/18 (MITECO Síntesis de los planes hidrológicos españoles. Segundo Ciclo de la DMA (2025-2021 and own elaboration).

Figura 15. Comparación entre las escorrentías de las Demarcaciones Hidrográficas, serie 1940/41, 1979/80 y serie 1980/81 – 2017/18 (MITECO Síntesis de los planes hidrológicos españoles. Segundo Ciclo de la DMA (2025-2021 y elaboración propia).

The new hydrological plans of the third cycle reflect as an objective an effort to reduce water demand. This can be seen in Figure 16, which shows how the water consumption targets in the new plans are significantly lower than the forecasts made in the plans of the Second Cycle. Adaptation to climate change in certain sectors, such as irrigation, could mean an increase in water needs that must be compensated with a new crop orientation, the improvement and modernisation of hydraulic infrastructures (catchment, transport, etc.), modern water application techniques on plots (sprinkling, drip, etc.), efficient water use, and support technologies for saving and efficient water use (weather forecasts, soil moisture sensors, recommendations for irrigation allocations, etc.).

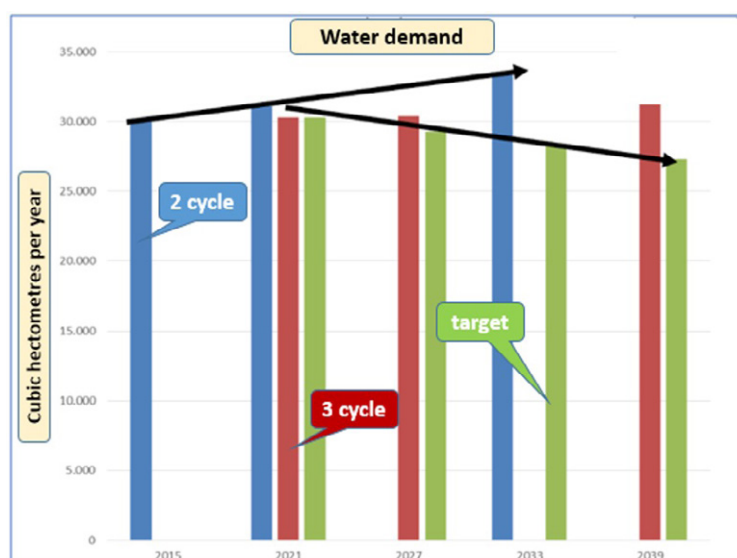


Figure 16. Water demands in the second (2016- 2021) and third cycle (2022-2027) plans and target values (River Basin Management Plans/Planes Hidrológicos 2016-2021 y 2022-2027 and own elaboration).

Figura 16. Demandas de agua en los planes del segundo (2016- 2021) y tercer ciclo (2022-2027) y valores objetivo (Planes Hidrológicos 2016-2021 y 2022-2027 y elaboración propia).

4.3.2. Third planning cycle (2022-2027) Status of water bodies and environmental objectives

The percentages of water bodies that reached the environmental objective of good status in 2015 foreseen in the first cycle (2010-2015) were lower than those established in the second cycle plans (2016-2021). And the same is true (year 2021) between those established in the second cycle and those verified in the new plans of the third cycle (2022-2027) (Fig. 17).

Therefore, a major effort will be required in the latter period to achieve the ambitious targets proposed. This implies both investments in new infrastructures and the improvement of existing ones, including wastewater treatment, as well as management measures for an adequate control of discharges.

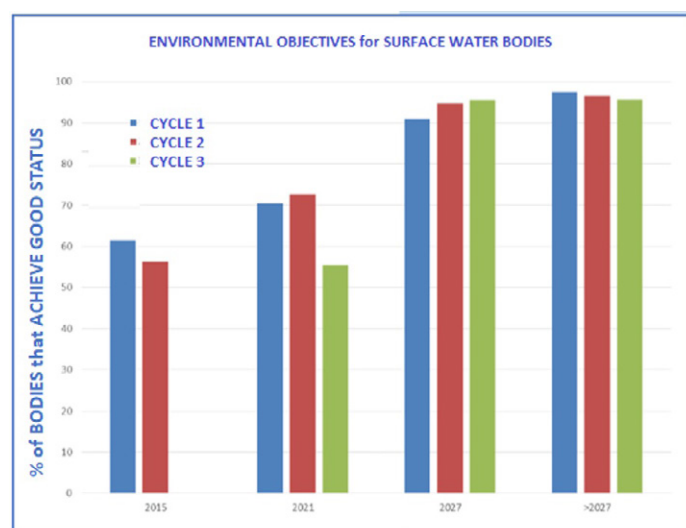


Figure 17. Environmental objectives on surface water bodies: Percentage of surface water bodies achieving good status and environmental objectives (River Basin Management Plans/Planes Hidrológicos).

Figura 17. Objetivos ambientales sobre masas de aguas superficiales: Porcentaje de masas de agua superficial que alcanzan el buen estado y objetivos ambientales (Planes Hidrológicos).

A similar situation can be seen in the status of groundwater bodies and the proposed environmental objectives (Figure 18). In this case, there is the added operational difficulty in the application and control of the codes of good agricultural practices by the corresponding responsible administrations. This slows down the fight against diffuse pollution, especially in areas declared vulnerable to nitrate pollution, and adequate cooperation between the administrations involved is necessary.

In many areas of the territory, the scarcity of the resource to meet water demands is one of the main pressures that put the environmental objectives established in the plans for the different bodies of water at risk of not being achieved. In these situations, the achievement of environmental objectives requires the establishment of restrictions on groundwater abstractions and surface water flow regimes.

In the first case, the aim is to avoid endangering the good quantitative status of groundwater bodies, in short, the overexploitation of aquifers.

In the case of surface waters, environmental flow regimes are established to contribute to achieving good ecological status or potential in rivers or transitional waters. Since the second cycle plans, the imposition of environmental flow regimes has been extended to almost all water bodies classified in the “rivers” category.

In this process, it is necessary to introduce more efficient management measures and, above all, the modernisation of irrigation systems, which has been promoted with determination in this century, in order to make economic uses of water compatible with the preservation of the ecosystems associated with water, and to guarantee the sustainability of the future scenario. As indicated above, it is essential to consider the role of water in energy sustainability and food sustainability.

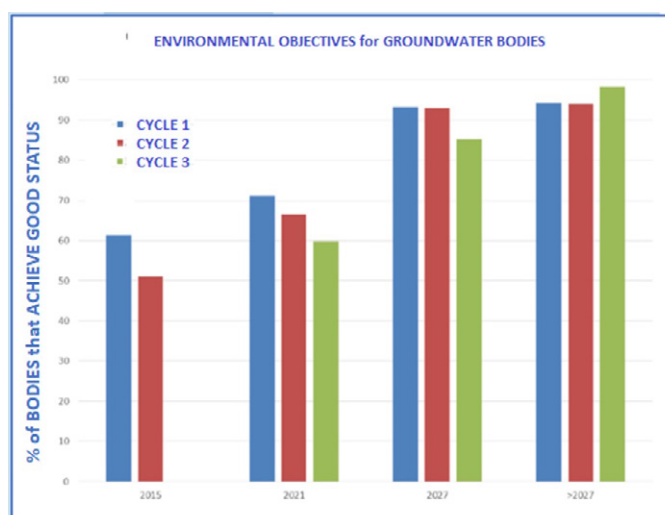


Figure 18. Environmental objectives on groundwater bodies: Percentage of groundwater bodies achieving good status and environmental objectives (River Basin Management Plans/Planes Hidrológicos).

Figura 18. Porcentaje de masas de agua superficial que alcanzan el buen estado y objetivos ambientales (Planes Hidrológicos).

4.3.3. Recovery of the cost of water services

In accordance with the provisions of the WFD and our water legislation, the plans must include a study on the recovery of the costs of water services. These consist of all services for the benefit of households, public institutions or any economic activity consisting of the abstraction, reservoir, storage, treatment, or distribution of surface water or groundwater, as well as the collection and treatment of wastewater that is subsequently discharged into surface water (*Dirección General del Agua del Ministerio para la Transición Ecológica y el Reto Demográfico (2016)*). Primera evaluación de la idoneidad de los instrumentos de recuperación del coste de los servicios del agua en España).

The new plans incorporate the environmental cost, or additional cost that has not been previously internalised, necessary to recover the good status of the bodies of water in order to compensate for the deterioration they may have suffered due to the introduction of the water service. This is an important factor for countries, such as Spain, where hydrological irregularity means that such services require significant investment in infrastructures to make these services feasible.

In the hydrological plans of the Spanish River Basin Districts, the valuation of the environmental costs is carried out considering the costs of the measures foreseen to achieve the environmental objectives.

Cost recovery rates (CRR) have been calculated on the value of financial costs and on total costs, including environmental costs.

A summary of the results is given in the *Table 3* and *Table 4*. The sum of the costs for each of the services considered in *Table 3* and the total costs are shown in *Table 4*.

A higher rate of cost recovery is observed in the use of groundwater, the use of which is carried out to a greater extent by private initiative, as opposed to the use of surface water, in which the presence of public intervention is greater and is subject to the application of legally regulated tariffs and charges to its beneficiaries. In the latter case, these are, in general, interventions that require actions on a larger scale and have a greater social effect.

The cost recovery analysis considers the levies (taxes, charges or other types of contributions) legally applicable to users of the works and services financed by the Public Administrations. *Table 5* shows the origin of these revenues and the water services whose costs they contribute to recovering.

Table 3. Costs of water services and cost recovery rates (CRI) plans 2016-2021.

Source: River Basin Management Plans/Planes Hidrológicos.

Tabla 3. Costes de los servicios del agua e índices de recuperación de costes (IRC) planes 2016-2021.

Fuente: Planes Hidrológicos.

Water Services		Water use		Total financial cost (M€/y)	Environmental cost (M€/y)	Income (M€/y)	Financial RC Index %	Index RC total %
				A	B	E	E/A	E/A+B
Extraction, storage, treatment and distribution of surface and groundwater	1 High-pressure surface water services	1 Urban		464.26	116.52	329.59	71%	57%
		2 Agriculture/Livestock		393.47	106.79	243.71	62%	49%
		3 Industry/En ergy		174.39	62.48	138.11	79%	58%
	2 High-pressure ground water	1 Urban		181.65	11.778	162.96	90%	84%
		2 Agriculture/Livestock		114.38	21.90	99.49	87%	73%
		3 Industry/En ergy		14.37	0.26	12.08	84%	83%
	3 Water distribution for low-pressure irrigation	1 Agriculture		911.45	238.80	550.24	60%	48%
		1 Homes		2,551.2	29.99	2,143.3	84%	83%
		2 Agriculture/Livestock		52.92	6.47	36.71	69%	62%
	4 Urban supply	3 Industry/En ergy		559.27	6.72	445.53	80%	79%
		1 Homes		521.29	78.03	504.40	97%	84%
		2 Agriculture/Livestock		1,381.0	366.69	1,313.8	95%	75%
	5 Self-service	3 Industry/En ergy		336.1	59.04	310.84	92%	79%
		1 Homes		14.93	1.20	13.74	92%	85%
		2 Agriculture/Livestock		32.36	8.84	9.53	29%	23%
	6 Reuse	3 Industry/En ergy		12.17	1.90	11.75	97%	83%
		1 Homes		77.36	5.40	50.80	66%	61%
		2 Agriculture/Livestock		58.78	3.40	44.62	76%	72%
	7 Desalination	3 Industry/En ergy		12.61	0.20	6.11	48%	48%
Collection and treatment of discharges into surface waters	8 Collection and purification outside public networks	1 Homes		47.16	9.88	38.92	83%	68%
		2 Agriculture/Livestock		5.66	7.05	4.83	85%	38%
		3 Industry/En ergy		200.60	28.50	179.23	89%	78%
	9 Collection and purification of public networks	1 Urban		2,093.0	548.98	1,530.4	73%	58%
		3 Industry/En ergy		552.74	138.73	396.26	71%	57%

Table 4. Total cost values of water services and their recovery (2016-2021 plans).

Source: River Basin Management Plans/Planes Hidrológicos.

Tabla 4. Valores totales de costes de los servicios del agua y de su recuperación (planes 2016-2021).

Fuente: Planes Hidrológicos.

Services	Costs			Income	Cost recovery	
	Financial	Environ- mental	TOTAL		Financial IRCs	Total IRC
High-pressure surface water services	1,032.12	285.79	1,317.9	711.4	69%	54%
High-pressure groundwater	310.41	39.94	344.34	274.53	88%	80%
Water distribution for low-pressure irrigation	911.55	328.80	1,150.35	550.24	60%	48%
Urban supply	3,163.38	43.18	3,206.56	2,625.57	83%	82%
Self-service	2,238.40	503.76	2,742.16	2,129.11	95%	78%
Reuse	59.46	11.94	71.40	35.01	59%	49%
Desalination	148.75	9.00	157.75	101.53	68%	64%
Collection and purification outside public networks	253.42	45.43	298.85	222.98	88%	75%
Collection and purification of public networks	2,645.74	687.71	3,333.45	1,924.69	73%	58%
TOTAL	10,763.23	1,859.56	12,622.9	8,575.07	80%	68%

Table 5. Types of charges for the use of infrastructure and services financed by public institutions, which count towards the recovery of costs of water services. SSEE, Sociedades Estatales; SEISA Sociedad Mercantil Estatal de Infraestructuras Agrarias S.A; Canon of use DPH is the fee for the use of the public water domain.

Tabla 5. Tipos de exacciones por la utilización de infraestructuras y servicios financiados por instituciones públicas, que computan en la recuperación de los costes de los servicios de agua. SSEE, Sociedades Estatales; SEISA Sociedad Mercantil Estatal de Infraestructuras Agrarias S.A; Canon de uso DPH es la tasa por el uso de agua de dominio público.

Water Service		Regulation fee	Tariff for water use	SSEE: Water supply and sanitation	SEISA tariff	Tariffs for water transfers	Changes for upstream operators	Irrigation tariffs	Urban distribution tariffs	Reuse	Desalination	Sewerage tax	Discharge control fee	Sewerage charges	Tax on discharges into coastal waters	Infringements for damage to the DPH	Hydroelectric charges	DPH usage fee	Hydromorphological taxes	Water charges in Catalonia
Extraction, impoundment, storage, treatment and distribution of surface water and groundwater	1 Upstream surface water services																			
	2 High groundwater services																			
	3 Downstream irrigarian water distribution																			
	4 Urban water supply																			
	5 Self-service																			
	6 Reuse																			
	7 Desalination																			
Collection and treatment of discharges to surface waters	8 Collection and treatment outside public networks																			
	9 Collection and treatment outside public networks																			

4.3.4. Programme of measures

The programmes of measures of the plans currently in force (Spanish Hydrological Plans of the River Basin Districts 2022-2027. PPHH 2022-2027) include the set of actions necessary to achieve the environmental objectives, as well as other objectives related to Spain's hydrological singularity in the European context.

They cover the following groups of measures:

- Measures required by the WFD aimed at achieving environmental objectives
- Investments to improve the supply of resources aimed at meeting demand
- Measures to mitigate the effects of extreme hydro-meteorological events (droughts and floods).
- Governance and knowledge improvement measures

Other investments required by the various water-related uses Table 6 provides a summary of the estimated investments required to develop the planned actions and measures.

Table 6. Investments contemplated in the Programmes of Measures of the PPHH 2022 – 2027.

Source River Basin Management Plans/Planes Hidrológicos 2022-2027.

Tabla 6. Inversiones contempladas en los Programas de Medidas de los PPHH 2022 – 2027.

Fuente: Planes Hidrológicos 2022-2027.

	investments AGE (mill.€)	Total investments (mill.€)	Sum of investments 2022-2027 (mill.€)
1. General and hydrological planning studies	228.666	476.55	426.71
2. Management and administration of the public water domain	482.722	1,998.86	1,358.356
3. Hydrological monitoring and information networks	425.391	769.12	611.02
4. Restoration and conservation of the public water domain	875.865	2,112.89	1,656.17
5. Flood risk management	1,372.107	3,292.07	2,599.34
6. Infrastructures			
6.1 Regulation infrastructures	479.691	1,494.36	633.45
6.2 Irrigation infrastructures	1,968.706	7,466.67	5,441.24
6.3 drainage and purificarian infrastructures	3,293.137	12,813.39	10,950.57
6.4 Supply infrastructures	599.938	4,369.83	3,137.48
6.5 Desalination infrastructures	874.528	1,498.60	1,364.48
6.6 Reuse infrastructures	204.778	1,076.22	836.59
6.7 Other infrastructures	1,009.576	4,106.63	3,331.68
6.8 Infrastructure maintenance	618.935	2,045.77	1,649.30
7. Security of infrastructures	264.939	738.56	445.90
8. Aquifer recovery	421.808	891.25	848.95
9. Other investments	332.707	2,081.48	1,500.93
Grand total	13,454.372	47,259.40	36,812.49

Among these actions, the following can be highlighted for their special relevance and uniqueness:

- Those aimed at modernising and improving hydraulic infrastructures for irrigation (reduction of water losses in canals and pipes, change from traditional irrigation systems to sprinkler or drip systems, etc.). This measure is congruent with the path set out in the plans for the moderation of water demand, of which irrigation is the largest consumer. In Spain, irrigation is key to the agri-food system, especially for fruit and vegetable production and extra-early crops, sectors in which our country is highly competitive in the European market.
- Those related to the revision of Directive 91/271/EEC on urban waste water treatment, with its amendments on urban waste water collection, treatment and discharge systems and the protection of waters against untreated discharges.

Compliance with this budget chapter is essential for making progress in achieving good status of water bodies, an objective on which Spain is notably behind schedule and for which the European Union has several investigation and sanction proceedings. In this area, previous specific plans provided for the following investments:

- National Sewerage and Waste- Water Treatment Plan (1995) 12,000 M€.
- National Water Quality Plan 18,000 M€.

These plans were partially implemented. The River Basin Management Plans (Planes Hidrológicos) of the River Basin Districts PPHH 2022-2027, set a total investment of 12,180 M€ (of which 3,293 M€ are to be carried out by the National Government) with 10,950 M€ corresponding to the period 2022-2027. In addition, an additional €7 billion is estimated for the implementation of the new Urban Waste - Water Treatment Directive.

- Actions regarding the safety of dams and their reservoirs, required by the national regulations on this matter (RD 264/2021), in view of the extensive dam heritage existing in our country (more than 1,100 large dams, with a capacity of more than 61,000 hm³ of capacity) which historically had to be built to obtain sufficient water availability in an irregular hydrological regime, unparalleled in the European context. The age of many of these works and the accumulated deficit of investment in their proper conservation justifies the importance of this chapter, which must be reinforced. These actions have been summarised in the Priority Lines of Action on dam safety 2023-2033, which envisage the need to invest 4,200 million euros over the next 10 years to conserve the functionality of this great hydraulic heritage on which water security in our country depends.
- In addition to the investments contemplated in these Hydrological Plans, the investments necessary to achieve the decarbonisation objectives of the electricity sector are also particularly relevant, in the context of the roadmap of climate measures for the reduction of GHG emissions established after the Paris Agreements (2015). The National Integrated Energy and Climate Plan (PNIEC) established national objectives to be achieved by 2030 and to pursue the complete decarbonisation of the electrical system by 2050, promoting renewable generation systems.

To achieve the integration of photovoltaic, solar thermal, wind power plants, etc. production into the system, the 2023 revision of the PNIEC, recently approved by the government and submitted to the EU this summer, has provided for the expansion of hydro storage capacity to reach a capacity of 18.9 GW by 2030, of which 10 GW are seasonal storage, and will mainly reside in hydro energy storage.

This has led to the appropriate legal amendment in 2023 to give priority to reversible hydro plants in the processing of the concessions of hydropower plants. This hydro energy storage is expected to mobilise private investment in the order of + 40 billion euros.

- These River Basin Management Plans (Planes Hidrológicos PPHH 2022-2027), on the other hand, do not incorporate the specific measures envisaged in the Flood Risk Management Plans, which in the second cycle involve more than 2.5 billion euros.

The greatest effort of these measures is concentrated on sanitation and wastewater treatment, with a clear presence of actions linked to the water-food nexus. And investments in dam safety and in the water-energy nexus should be reinforced.

In accordance with the current distribution of competences (in accordance with the constitutional framework and the provisions of our legal system) a very important part of the investments foreseen in the Programmes of Measures of PPHH 2022-2027 of the River Basin Districts (Fig. 19), corresponds to the Regional Autonomous Communities and Local Administrations, particularly in the field of urban supply, sewerage and wastewater treatment. Therefore, the implementation of these measures requires a considerable effort in institutional co-ordination and co-operation.

These programmes are very ambitious and require the mobilisation of significant budgetary resources, which will only be available if European funds (Next Generation, ERDF, etc.) are applied and if the cost of water services is increased. Given the necessary support to be obtained in water tariffs, a specific study is needed on its financing (sources of money and quantification of the amount eligible for subsidies in each case), prioritisation of actions and how to promote effective public-private and inter-administrative collaboration.

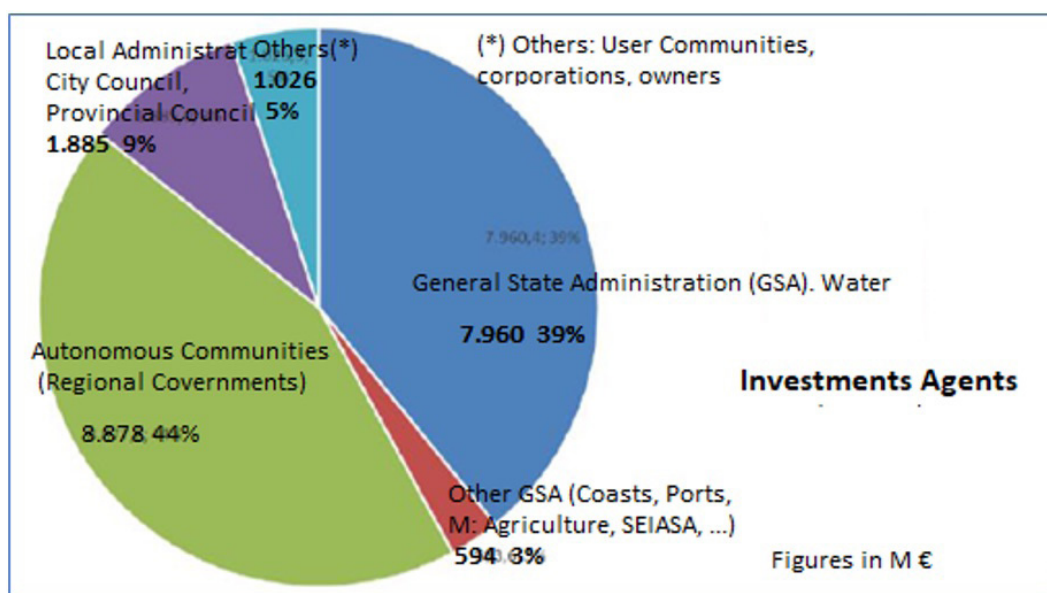


Figure 19. Participation of the agents responsible for the distribution of investments in the Programmes of Measures of the Hydrological Plans of the River Basin Districts.

Figura 19. Participación de los agentes responsables en la distribución de inversiones de los Programas de Medidas de los Planes Hidrológicos de las Demarcaciones hidrográficas.

5. WATER CHALLENGES

Based on the above, it is easy to deduce that it is not enough to follow an inertial trend in order to face the challenges we face so that water continues to fulfil its vital fusion for sustainable development and does not become a critical resource that strangles the wellbeing of our society. These challenges and some ideas for addressing them, from a personal perspective and in the light of our extensive professional experience, are set out below.

5.1. Water belongs to all and for all

Water is a resource that is part of a public water domain that must be managed and used as such. We are sadly witnessing an escalation in which water is used not as an element of understanding between territories, but as a tool for the affirmation of some autonomous regions against others, introducing political barriers to proper river basin management, and limited interconnection actions between them, in order to solve problems that cannot be solved within a river basin.

This means betraying our being and our rich hydraulic heritage, in which up to now, fortunately, public action has followed an orientation dictated by satisfying objectives and not stopping at administrative frontiers. In this sense, the keys that we believe should be followed in the future are as follows:

- Only a national policy can promote consensus, build bridges and avoid confrontation.
- Users are key to this. They must be listened to, understood and supported.
- Water is key in a country like Spain, due to the spatial and temporal irregularity of water resources.
- The European Union finds it difficult to understand us. There is a great need to develop an active policy towards the EU at the national level so that the directives do not take us away from sustainable competitiveness.

5.2. Water, a matter of state

Closely related to the above. It is necessary to establish a water policy that raises the level of action, highlights it as a priority issue on the political agenda and as such allocates renewed efforts and means to solve the problems that will arise if lukewarmness continues to permeate our action. In this sense, the keys that we believe should be followed in the future are these:

- Do not make water a matter of conflict and dissent, but of consensus among all.
- Promote a National Water Pact, political, with users and social agents, that is sustained over time and provides stability to the necessary actions.
- Approve a new National Hydrological Plan supported by an updated Water White Paper and based on the National Water Pact.
- Strengthen governance and revitalise water administration, which cannot continue to be a Directorate General in the organisation chart of the Ministries, nor languish for lack of resources in the basin organisations. Avoiding that the announcements of a review of the Confederaciones Hidrográficas (river basin organizations, dependent on the General Administration) hide a desire to take away their capacity for action in order to strengthen the Regional Autonomous Communities in their place.

5.3. Water security and climate change resilience

Spain does not escape the vectors that move the world, and the water security that the work of our ancestors has made a reality and which we enjoy is being threatened as a result of the obsolescence of our hydraulic works (especially dams, we are “reservoir-dependent”) and the necessary adaptation to climate change. In this sense, the keys that we believe should be followed in the future are as follows:

- Launch a new Dam Safety Programme, which requires training professionals and dedicating economic resources, following the Priority Lines of Action already formulated by the Directorate General for Water.
- Pay due attention to the dams and large state-owned pipelines, which represent a great heritage that, due to their age, must be replaced and maintained in good condition: do not lose their heritage value, dedicate funds and means so that they can provide service in good condition. Do not live off rents and decapitalise
- Advance in the application of digitalisation and technology to optimise their exploitation.
- Decision support systems for the management of:
 - Operating systems
 - Flood management and flood protection
- Interconnection of systems and new regulations for climate change adaptation
- Promote sustainable development in basins that still have resources available in certain areas (Duero, Ebro, Guadiana-Tinto, Odiel and Piedras...).
- Activation and integration of non-conventional resources (desalination and reuse)
- Support and maintain a Groundwater Action Plan.
- Support a sustainable urban water cycle, at an appropriate scale, with a national level regulator to identify inefficiencies and disseminate good practice.

5.4. Setting up integrated water systems

In several Spanish basins, and especially in the southeast of Spain, the situation of water stress hinders the effective and efficient management of the available resource with respect to the environmental limits defined in the hydrological planning. This situation will be aggravated by a foreseeable reduction in available water resources in the short term (both external resources from the Tajo-Segura water transfer or the Negratín-Almanzora pipeline, and the resources of these Mediterranean districts as a result of the reduction in natural contributions associated with climate change), as well as the existence of various bodies managing the hydraulic infrastructures and available water resources.

The response to this problem requires the joint, more effective and more efficient use of conventional and non-conventional resources (desalination and reuse), bearing in mind that the production capacity of these non-conventional resources is under-utilised due to the lack of interconnection capacity between storage facilities and connection to demand points, the high cost of such use, and the bureaucratic problems associated with the allocation of these resources.

In short, we must seek to design an organisational architecture that strengthens the resilient and sustainable governance of our water management model and its capacity to adapt to climate change.

To this end, it is proposed as a solution to this situation the creation and implementation of what has been called “Integrated Water Systems”, an organisational, legal and economic model for managing the supply and demand of water resources, the main defining elements of which would be the following:

- The coordinated and interconnected management of the different existing infrastructures linked to the management of water resources, both conventional (reservoirs, water transfers, pipelines, storage facilities, wells) and non-conventional (desalination and reuse facilities), maintaining the individual management of each infrastructure by its current owner.
- The incorporation of all the water resources from the different sources, decoupling the management of these resources from the corresponding infrastructures. Each operator continues to carry out the current function, but the resource is managed in a unified manner by the basin organisation, optimising this management from the technical, economic, water quality, energy and environmental points of view. The integration of non-conventional resources with traditional ones is facilitated, and they are managed like electricity, with a pool of resources managed by the administration, and the rights to use water or water concessions are given from the pool.

An important aspect for the integration of all resources from different sources is the special consideration of groundwater in this process. To achieve this, schemes for the joint use of surface water and groundwater must be established to optimize their overall use, taking advantage of the storage capacity and ubiquity of the aquifers where they are found.

- Allocation of the available water among the different users, in the most appropriate and convenient way for the integrated system as a whole, taking into account the total available resources and the existing demands, as provided for in the applicable hydrological planning in force.
- Apply decision support systems to optimise integrated water management in them (a pilot is underway in the Centro para el Desarrollo Tecnológico CDTI del Ministerio de Ciencia e Innovación, for the Segura basin).
- Apply harmonised, fair and equitable tariffs for the supply of available water resources that are affordable but consistent with the cost recovery principle of the Water Framework Directive, without squeezing out existing users.

5.5. Enhancing synergies in the WATER-FOOD-ENERGY nexus

Our country offers some very favourable elements for it to be a reference in this matter: we have sun, land, water, topographical differences and experienced people. So the keys that we believe should be followed in the future are these:

- Strengthening the agri-food system as a key element of our economy and for EU food security.
- Consider irrigation as an element of territorial structuring and face the demographic challenge.
- Complete the modernisation of irrigation.
- Develop a National Hydraulic Energy Storage Program (NHESP), already considered in the 2023 revision of the PNIEC, taking advantage of the State's dams and reservoirs, and accelerate the processing and construction of reversible hydroelectric CHR plants by the private sector, to make the integration of non-manageable renewable energies (wind power plants and PV) a reality.
- Energy for water: use of the upper reservoirs of the RHCs (e.g., canal on the right bank of the Ebro on a counter-slope) and of non-manageable renewable energy discharges to provide cheap energy for hydraulic works (pumping, upper reservoirs for pressurised irrigation, new regulation reservoirs in diversion, water reuse systems, desalination...).
- Reversing the benefits of hydropower developments in the areas where they are implemented in order to meet the demographic challenge in those areas.

5.6. Water, a pillar of environmental policy

Having a resource but not having the right quality is of no use. Not achieving good status of water bodies. In addition, water provides important services for associated ecosystems. There will be no sustainability if these environmental issues are not addressed and resolved.

- Harmonisation and effective and realistic implementation of an adequate and sustainable regime of ecological flows. Reconsider them with homogeneous criteria throughout Spain.
- Maintaining the National River Restoration Strategy
- Develop regeneration plans for emblematic natural areas associated with water (Ebro Delta, Marismas de Santoña, Albufera de Valencia, Mar Menor, Doñana, Tablas de Daimiel, etc.).
- Launch economic signals to contribute to a better use of water and the preservation of associated ecosystems: consider introducing a water extraction and use control fee to incorporate the cost of the resource, both for surface water and groundwater, as well as desalinated and reused water.
- Launch a new National Waste - Water Treatment Plan, solving the problems of investment in sanitation and wastewater treatment. Defining not only the necessary actions, but also the issues related to sources of financing, the amount of European or national public aid, the scope of the repercussion of costs, the prioritisation of actions, the actors that have to develop them, and the possibilities of public-private collaboration.

6. WHY INVEST IN WATER

Evidently, many of the actions foreseen in the programmes of measures of the plans obey the fulfilment of regulatory obligations derived from our membership of the EU and the specialisations of our own regulatory framework in order to respond to the hydrological singularities that differentiate us from the European context.

This alone would serve to justify the public investment required and to drag in the private investment needed to avoid the penalties that might result from non-compliance.

However, there are sufficient reasons for the profitability of investments in the water sector to interest the private sector and, in particular, the development of public-private partnerships.

And, above all, it should be stressed that investments in water (essential for life and productive activities) are a commitment to the future, with the aim of guaranteeing sustainability in the use of this precious resource, and the best indicator that its use does not exceed the barrier of over-exploitation with irreversible effects is the conservation of the good condition of bodies of water and their associated ecosystems. (World Water Assessment Programme WWAP 2015 United Nations World Water Development Report Water for a Sustainable World. Paris, UNESCO Publishing/ Imprimerie Centrale S. A.). (World Water Assessment Programme). WWAP 2021. United Nations World Water Development Report Valuing Water. Paris, UNESCO Publishing/ Imprim. Centrale S. A.)

6.1. Social and economic profitability of investments in water and sanitation.

Firstly, and as a reference for contextualisation in our environment, annual investment in water in Spain barely reaches 0.11% of GDP, while in countries such as France, Germany, Italy and the United Kingdom it is around 0.25% of GDP. The Development Bank for Latin America (CAF) recommends that these investments should be around 0.3% of GDP for the first installation and 0.5% of GDP for operation and maintenance.

All these references make it advisable to increase the rate of investment in Spain to enable better conservation and management of our hydraulic heritage.

First of all, the importance of investments to ensure an adequate water supply and sanitation system must be emphasized.

As an example, various experiences and conclusions from international organizations in the EU and the rest of the world can be added. So, the World Meteorological Organisation (WMO) estimates that the value of the benefit obtained is 3 to 34 times the value of the budget invested in drinking water and sanitation, in addition to the improvement in the lives of millions of people.

Similarly, studies by the World Bank (2008) in Southeast Asia on the consequences of underinvestment in water and sanitation concluded that (for that reason alone) they resulted in associated economic losses of 7.2 % of GDP in Cambodia and 5.4 % of GDP in Laos. Similar studies by the Bank in 2008 estimated that the health costs directly attributable to poor water and sanitation services in Ghana, or by other institutions such as OECD (Organisation for Economic Co-operation and Development 2010. Financing Water and Sanitation in Developing Countries: The Contribution of External Aid. Paris, OECD) or Stockholm International Water Institute (SIWI 2005. Making Water a Part of Economic Development: The Economic Benefits of Improved Water Management and Services. Stockholm, SIWI). And in other specific works of the World Bank (2012 Achieving financial sustainability and recovering costs in bank financed water supply and sanitation and irrigation projects. Alexander McPhail, Alain R. Locussol, Chris Perry).

The work of a wide variety of authors confirms the positive effects of investments in drinking water on the economy, health and social well-being in recipient countries, such as Vahala, R. (2004). European Vision for Water Supply and Sanitation in 2030. Water Supply and Sanitation Technology Platform. And Vairavamorthy, K. (2008). Cities of the Future and Urban Water Management. Paper presented on 27 June 2008 during Thematic Week 2 of the Zaragoza International Exhibition, 2008.

These issues have been the subject of special interest by World Health Organization (WHO), and have been dealt with in a special way (WHO 2004. Evaluation of the Costs and Benefits of Water and Sanitation Improvements at the Global Level. Geneva; WHO 2006. Meeting the MDG Drinking Water and Sanitation Target: The Urban and Rural Challenge of the Decade. Geneva/New York; WHO . (2008). The Global Burden of Disease: Update. Geneva; WHO/UNICEF. (2010). Joint Monitoring Programme for Water Supply and Sanitation. Progress on Sanitation and Drinking-Water: Update. Geneva/New York; WHO/UNICEF. (2012). Joint Monitoring Programme for Water Supply and Sanitation. Progress on Sanitation and Drinking-Water: 2010 Update. Geneva/New York.

On the other hand, the goals for achieving Goal 6 of the 2030 Agenda will require the development of infrastructure and other measures to achieve universal access to safe drinking water and sanitation, improve water quality by reducing pollution, and increase the efficient use of water resources, justifying the need for significant investments in the water sector (WFEO-World Federation of Engineering Organizations reports (2022): Achieving SDG 6 (water goals): contribution from engineering)

Undoubtedly, the investments to be made in water extend to many other fields besides supply and sanitation, which would require a more exhaustive analysis than the space devoted to this article. In some way, some aspects have already been mentioned in this document in relation to the effects of CC on water resources (impact on the availability of water for consumption, frequency and intensity of droughts, increase in the torrential nature of rainfall and effects on floods, etc.), the relevance of the water-energy binomial and the water-energy-food trinomial, etc. (WWAP World Water Assessment Programme, 2014). United Nations World Water Development Report Water and Energy. Paris, UNESCO Publishing/ Imprimerie Centrale S. A.)

In relation to investments to avoid or mitigate the effects of hydrological hazards, studies in the United States estimate the payback value of these investments to be in the order of 1 to 6. This should guarantee the profitability of the investments to be developed in this field (WFEO-World Federation of Engineering Organizations reports 2019. Best practices on drought and flood management: Engineer's contribution)

Investments in adaptation to climate change should be seen not as simple costs, but as profitable investments to the extent that they will avoid or reduce the damage derived from the negative effects predicted in the successive IPCC reports, repeatedly cited. (World Bank 2010. The Cost to Developing Countries of Adapting to Climate Change. New Methods and Estimates. The Global Report of the Economics of Adaptation to Climate Change Study. Consultation Draft. Washington D.C. and. the Economics of Adaptation to Climate Change. Background Papers: Costs of Adaptation Related to Industrial and Municipal Water Supply and Riverine Flood Protection P. J. Ward, P. Pauw, L. M. Brander, Jeroen, C. J. H. Aerts and K. M. Strzepek). (World Federation of Engineering Organizations reports WFEO 2023. Adaptation to Climate Change: Water and Engineering) (World Water Assessment Programme WWAP 2020. United Nations World Water Development Report Water and Climate Change. Paris, UNESCO Publishing/ Imprimerie Centrale S. A.).

6.2. The benefits of regulation developed by dam reservoirs and contribution of groundwater

The socio-economic contribution of dams and reservoirs, and their contribution to the structuring of the territory, has not been specifically considered in the plans (PPHH 2022-2027), so there is a lack of a rigorous and global analysis of their influence on our country as a whole. However, there is no doubt about the great socio-economic effect derived from the regulation provided by these infrastructures, making it possible to develop a large number of activities, which, without the availability of water, would be unviable.

According to the White Paper on Water (Libro Blanco del Agua, 2000; Ministry of the Environment) (Figure 20), which analyses these aspects with technical rigour, the effect of the regulation of reservoirs has increased the availability of water for the different uses from 7% to 41% of the natural contribution (average value of annual renewable resources). This figure has not been detailed or updated in this third planning cycle, which could be obtained in each exploitation system and in each plan with the simulation models used (see the appendices of the balances and allocation of resources in the reports of the plans) and compiled in a synthesis report.

The overall figures given in the 2000 White Paper are as follows:

- Average natural inflow (average annual renewable resources). 110,116 hm³ /year
- Available in natural regime with variable demand: 7,112 hm³ /year (7%)
- Available regulated with variable demand: 39,175 hm³ /year (41%)

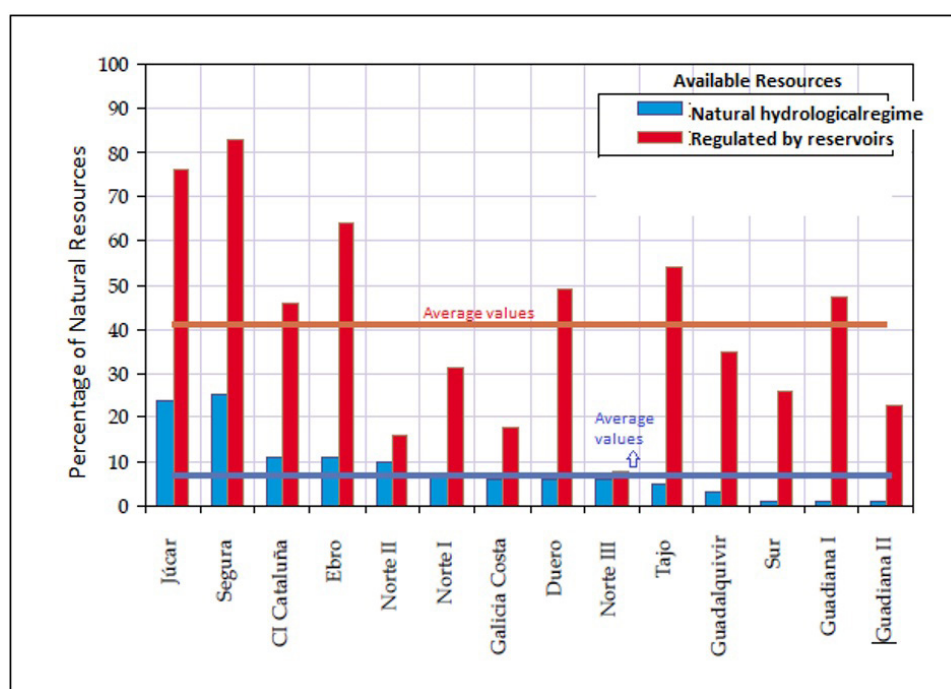


Figure 20. Resources available under natural and under surface water regulation regime by reservoirs (Figure 143 of the White Paper on Water, Ministry of Environment 2000).

Figura 20. Recursos disponibles en régimen natural y en régimen de regulación de aguas superficiales por los embalses (Figura 143 del Libro Blanco del Agua, M^o Medio Ambiente 2000).

To this increase in the availability of the resource, groundwater abstractions contribute additionally with some 6,547 hm³, although this amount includes overexploitation, estimated between 710 hm³ (Libro Blanco de las Aguas Subterráneas Ministerio de Industria y Energía, Ministerio de Obras Públicas, Transportes y Medio Ambiente, 1995) and 1,055 hm³ (Report 1993 on the proposed National Hydrological Plan. Ministerio de Obras Públicas, Transportes y Medio Ambiente)

There are no reliable figures on the value of water in Spain. However, approximations can be obtained. The White Paper (Libro Blanco del Agua en España. Ministerio de Medio Ambiente, 2000), while calling it into question, cites a Gross Value Added at market prices (GVAMP) of 2.45% of the productive sectors intrinsically linked to water. More recent studies, reflected in the Líneas de Actuación para la Mejora de la Seguridad de Presas y Embalses 2023-2033 (Ministerio para la Transición Ecológica y el Reto Demográfico, 2023), estimate that the water dammed by these infrastructures generates benefits to the Spanish economy of more than 35,000 M€ (≈ 2.9% of GDP).

Among the activities and sectors that have benefited most from the availability of water, achieved thanks to the regulation provided by these reservoirs and, to a lesser extent, by groundwater abstractions, the following can be highlighted:

- Hydropower production and Industry: The first two are economic activities that use water as a fundamental input in their production process, without which they would be unviable.
- Irrigation. Without reservoirs, no more than 1,000,000 ha of irrigated land (currently 4,043,923 ha are irrigated according to PPHH 2016-2021). The thesis is that these transformations are useful for achieving social results in terms of maintaining demographics, settling populations, territorial structuring and, in short, generating employment that makes rural life linked to irrigated agriculture sustainable. The decrease in population in irrigated areas has been clearly lower than in non-irrigated areas.
- Tourism and the service sector in general. The strong seasonality of tourism, concentrated in the months of August, July and September (an increase of more than 10% in these months).

The energy sector is the sector with the worst trade balance (imports minus exports) in Spain. The production of clean and renewable energy in hydroelectric dams helps to alleviate this marked deficit.

Tourism, on the other hand, provides Spain's main positive international trade balance.

The Spanish countryside has become highly productive and can generate a significant surplus in our balance of payments.

The food sector has a large foreign surplus (12,589 M€, 56% more than the automotive industry) and Spain has become a European food exporting power. The fruit, vegetables and legumes subsector were the fifth largest exporter in 2022, according to data from the Ministry of Industry, Trade and Tourism. Irrigated production accounted for 65% of final vegetable production. However, nothing is attributed to water in the national accounts.

It is difficult to quantify the "benefit" produced by hydrological regulation in the face of extreme phenomena (droughts and floods), which can be estimated by the value of the damage avoided or reduced during these episodes. It also improves the suitability of the territory for urban settlement and the development of economic activities, reducing - for example - its risk of flooding.

Most of the Special Action Plans for Situations of Alert or Possible Drought record as indicators the volumes of water stored in reservoirs and their deviation with respect to the usual ones for each month, as these reserves are what guarantee most of the water supplies demanded. In this section, it is important to highlight the importance of the strategic use of groundwater in a drought situation, through its transitory overexploitation during this period and its subsequent recovery once the hydrological crisis has been overcome. This operation is feasible thanks to the large volumes of water stored in the aquifers.

The management of dams and reservoirs during floods has taken an important qualitative leap forward through the incorporation of the most modern technologies in the development of the Automatic Hydrological Information System (AHIS). This complex system (Figure 21) allows the capture, in real time, of the most relevant hydro-meteorological data (rainfall, temperature, gauging of river flows, stored volumes and reservoir outflows, etc.), its remote transmission to control centres for processing and the application of hydrological models that allow forecasting of the hydrographs entering the reservoirs. The use of sophisticated decision support systems (DSS) offers a support tool for the operation of the dam's spillways and the evacuation of flows, so as to minimise flood damage downstream.

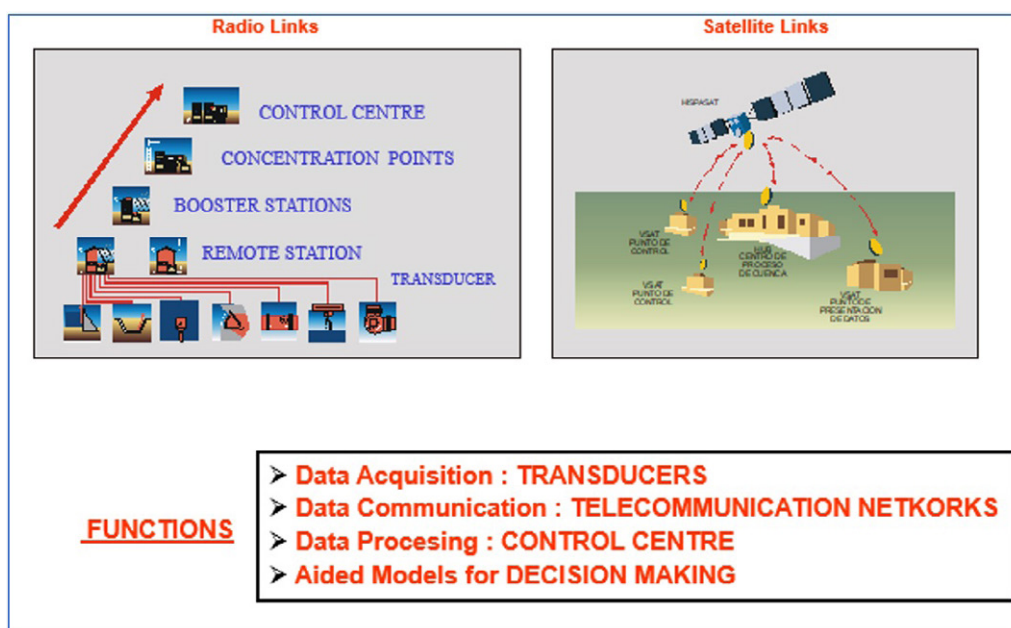


Figure 21. Automatic Hydrological Information System (AHIS). Communication systems and functions.

Figura 21. Sistema Automático de Información Hidrológica (SAIH). Sistemas de comunicación y funciones.

The AHIS is also a tool incorporated as a hydrological warning system in civil protection plans for flood risk.

The development of the AHIS has meant a great impulse in the technological advance of the operation of hydraulic infrastructures and water management, involving a wide sector (instrumentation, telecommunications, computer systems, engineering, research, etc.) and constitutes a paradigmatic example of the typology of future investments in relation to water.

The study carried out by the Directorate General for Water and the Guadalquivir Hydrographic Confederation on the Analysis of Damage Avoided during the episode caused by the Nelson squall, between 26 March and 7 April 2024, offers an illustrative testimony of the role played by the dams in this type of event. **Figure 22** shows the decrease in the peak flow of the floods.

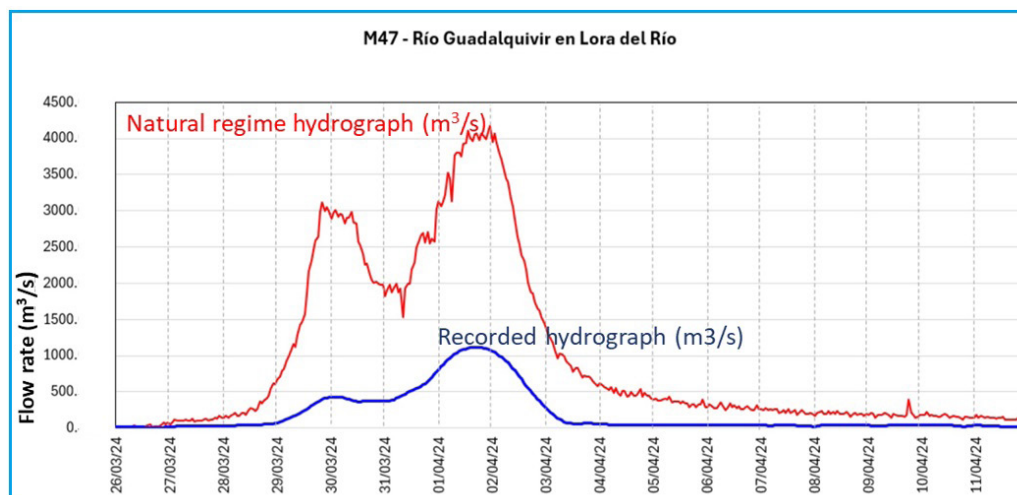


Figure 22. Comparison between the flood hydrograph that would have occurred in a natural regime and the one recorded in the Guadalquivir River (Lora del Río), during the Nelson squall. Source: SAIH of the Guadalquivir CH.

Figura 22. Comparación entre el hidrograma de avenida que se hubiera producido en régimen natural y el registrado en el río Guadalquivir (Lora del Río), durante la borrasca Nelson. Fuente: SAIH de la CH Guadalquivir.

The aforementioned study estimated the damage avoided at €6,160,541 and the number of inhabitants that would have been affected had the corresponding regulation infrastructures not been in place at 39,326.

In this same episode, and in the Guadiana basin, the damage avoided has been evaluated at 2,203,078,313 euros and 13,383 inhabitants who would have been affected if the corresponding regulation infrastructures had not been in place.

6.3. Importance and strategic value of groundwater

Throughout this article, the unique properties of groundwater have been repeatedly mentioned, highlighting their importance and strategic value, which cannot be sufficiently reflected through mere volumetric quantification of their use.

These circumstances justify the need for more specific and intensive attention from administrations and institutions. In particular:

- Programs to improve knowledge of the underground phase of the hydrological cycle: perimeter of aquifers, recharge, vulnerability, stored water reserves, hydrochemical...
- Improvement of water registry and inventory of groundwater withdrawals
- Improvement of extractions control
- Control of discharges
- Research and development programs.

6.4. It counts to invest in water and in the maintenance of its infrastructures

The previous paragraphs have summarised some of the actions necessary to carry out an adequate management of water resources and the investments foreseen for this purpose in the hydrological plans. However, a number of considerations and reflections can be added

The Asociación de Empresas de Conservación y Explotación de Infraestructuras (ACEX) deduces a value of water use, updated to 2023, of €5.9 billion. These values were obtained without considering other benefits such as, for example, the value of the flood abatement capacity and the reduction in the magnitude of a flood downstream.

The updated asset value of dams and reservoirs is €30,000 million. Discounting expropriations and environmental actions already carried out, 25,000 million euros are needed for their maintenance and operation. Considering a life cycle of 100 years, an annual figure of 250 M € would have to be dedicated to maintain this heritage in the proper conditions. (Priority Lines of Action Safety of Dams and Reservoirs 2023-2033: 4,400 M €).

It can therefore be concluded that reservoirs in Spain are very cheap, as the cost of their maintenance, between 250 - 440 M€/year, is more than compensated by their benefits, which provide a much higher value (multiplier higher than 20).

The Plans do not break down the costs of the services associated with the reservoirs, but they do detail the costs of high water and hydroelectric services. Synthesising the data from the different plans and with approximations using hypotheses that would require comparison, the recovery of financial costs is over 70%, and it should be taken into account that the general benefits (such as flood control) must be borne with taxes from all citizens.

There are other significant socio-territorial effects of reservoirs, some of which are of tax origin. Thus, for example, the IBI (Impuesto Bienes Inmuebles) and the IAE (Impuesto de Actividades Económicas) are directly passed on to the municipalities where they are implemented, maintaining the population in rural areas. Others, due to the effect they have on land use, allowing the development of urban and productive activities (industry, agriculture, leisure and tourism, etc.) in areas of the territory that would not be suitable for them (flooding, water shortages, etc.) if this type of infrastructure had not been implemented beforehand.

An important aspect that can be deduced from the River Basin Management Plans (called Planes Hidrológicos) of the River Basin Districts is the delay in achieving their environmental objectives, with many bodies of water failing to achieve good status or good potential status. This requires, among other things, a major investment effort in wastewater treatment and purification. This would save health costs, drinking water purification costs, improve the quality of life, and the situation of ecosystems associated with water resources, and improve the quality of life.

6.5. The role of States Companies (Sociedades Estatales) as a catalyst for public investment

State-owned companies (SEIASA, ACUAES, ACUAMED) play an important role in leveraging public resources and increasing management capacity to incorporate European funds, contributions from users and private financiers (European Investment Bank EIB). These act with statutory guarantees of transparency, allowing a more flexible and appropriate prioritisation of actions and formalising prior commitments between the parties.

The selection of the investor is made in each case by analysing the social interest of the intervention and the expected economic profitability of the action (Fig. 23).

When investments are fully or partially financed by State funds, because of their high social interest and uncertain profitability, water legislation provides for the application of a tariff regime (water use tariffs, regulation fees or other charges).

However, there are investments that are not recoverable by their nature, as the beneficiary is not a specific group but the general interest. Therefore, the general tariff regime cannot be applied. This is the case for flood prevention, environmental restoration of watercourses, etc.

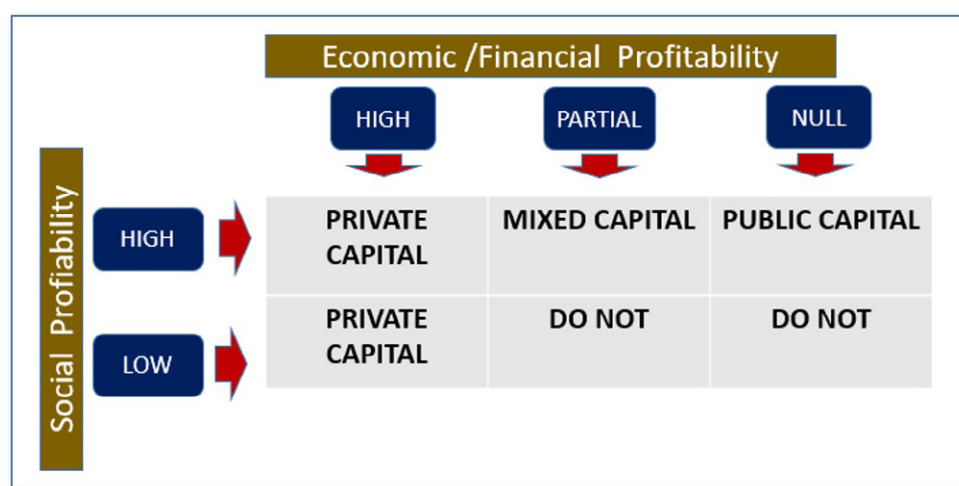


Figure 23. Decision-making scheme for the choice of investor and the financing of actions.

Figura 23. Esquema decisorio para la elección del agente inversor y la financiación de las actuaciones.

It seems necessary that this solid conceptual scheme be translated into a benefit-cost analysis of the various necessary actions, so that public money is used in a reasonable and justified way to achieve the general interest objectives set out in the water planning.

7. CONCLUSIONS

From the analysis carried out in previous sections, in relation to water resources in Spain in the global context, its main problems and challenges, the economic value of water and its importance in guaranteeing economic and environmental sustainability, the following conclusions can be summarised:

On the global context and principles for sustainable water resources management.

- Water is a critical global resource to ensure sustainable development of economic activity and quality of life. Its use must be framed in the context of human rights for the enjoyment of life in full dignity and equality.
- It is a renewable resource subject to the hydrological cycle. Groundwater and surface water are part of the same resource, through their atmospheric, surface and groundwater phases. Therefore, their management must be conceived in an integrated manner, through Integrated Water Resources Management (IWRM).
- The Agenda and the Sustainable Development Goals (SDGs) set the way forward in the global context. Water, the subject of specific targets in SDG 6 (clean water and sanitation), is interlinked in almost all the other goals, so water security affects all of them. Water is a crosscutting resource, not a separate sector.
- Special consideration should be given to the water - food - energy nexus.
- Climate Change (CC) is a global phenomenon on whose origin and consequences there is agreement in the scientific community (see Intergovernmental Panel on Climate Change (IPCC) reports). However, its effects cannot be precisely established on a local geographic scale, so its forecasting and management must be designed in a flexible and adaptive manner.
- In terms of water resource management, a considerable effort must be made in adaptive measures (efficiency, reduction of consumption...) that serve to combat the effects of CC impact. But it is also necessary to actively participate in mitigation measures, contributing to following the roadmap established after the Paris Agreements (2015) for the reduction of Greenhouse Gases (GHG) emissions.

- On water in Spain and singularities of the hydrological regime:
- The orographic and hydrological characteristics of the Spanish territory make it highly suitable for hydraulic storage, through the promotion of reversible hydroelectric plants that facilitate the integration of non-manageable renewable energies (wind power plants, photovoltaic, solar thermal, etc.) in the electricity sector. The necessary investments must be made to achieve the decarbonisation objectives defined in the National Integrated Energy and Climate Plan (PNIEC). This plan establishes the national objectives to be achieved by 2030, setting the milestones to achieve the complete decarbonisation of the electricity system by 2050.
- Transboundary watercourses should be the subject of international cooperation. EU water law (see WFD) - where applicable - and existing international treaties (Convention on the Protection and Use of Transboundary Watercourses and International Lakes / Helsinki Convention 1992) offer tools for their implementation on a bilateral or multilateral basis.
- The Spanish hydrological regime is characterised by its irregular distribution over time and territory (temporal and spatial irregularity), in comparison with Europe as a whole.

On the other hand, most of the consumptive water demand (80%) is dedicated to irrigation (4,000,000 ha in total) and is accentuated in the driest months, when the increase in urban consumption due to tourism is also concentrated.

The peculiar asymmetry between the competing demands for water and the natural supply of the resource to satisfy them has led to the need to build a large number of regulation infrastructures that would make it possible to guarantee water supply even in the months with the highest demand and lowest resources. As a result, the availability of water in Spain has required a greater investment effort than in the rest of Europe. And, therefore, a greater number of heavily modified or artificial bodies of water.

- Most of the resources available to ensure water supply come from regulated surface resources (over 72%, through more than 1,100 large dams), whose exploitation, conservation and maintenance is severely under-resourced.

The benefits of these infrastructures have not been rigorously assessed to date, and are not expressly included in the costs for water services estimated in the Spanish hydrological plans of the River Basin Districts. In the valuation of these benefits, in addition to those associated with agricultural production (livestock and agriculture), industrial and urban development, indirect benefits from the avoidance or reduction of damage caused by floods and droughts (extreme hydrological phenomena) should also be considered.

- The contribution of groundwater (with 21% of the total) is irreplaceable to complete the available resources necessary to guarantee urban water supplies, irrigation and industry. The ubiquity of the resource (see the map of groundwater bodies declared in the current hydrological plans of the Hydrographical Districts 2022 -2027) and the large reserves stored in their respective aquifers, confer a special aptitude for the use of the underground phase of the hydrological cycle in the provision of the water resource, with great economy of the infrastructures involved, and for its strategic use during periods of drought.
- Although non-conventional resources are much smaller, in comparative figures, they play a transcendental role in water security, contributing 398 hm³ from the reuse of treated wastewater and 658 hm³ from desalination, mainly from seawater. Spain occupies a prominent position in the world in the application of these technologies (leader in the EU in terms of volume reused).

The transfers between river basins of different River Basin Districts, with some 800 hm³ exchanged on average per year, contribute to balancing the unequal hydrological situation between territories.

On water planning and programmes of measures

- Planning is essential for the proper management and protection of water resources, as they require the anticipatory and preventive adoption of actions, the implementation of which requires long periods.

The last three planning cycles (2010-2015; 2016- 2021; and the current one, 2022-2027) - developed after the approval of the WFD - included their respective programmes of measures to achieve the environmental objectives set out in the plans and other specific goals of our planning. But there is a significant deviation between the investments foreseen in the programmes and those actually executed (barely a quarter of the investment to be made until 2021 had been executed by the end of 2020).

- The investments foreseen in the programmes of measures involve different actors. Only 30% correspond to the Public Water Administration of the General State Administration. It is essential to adopt organisational and regulatory actions that allow the necessary cooperation between the different competent administrations with responsibilities in water management.
- There has been repeated non-compliance with the objectives set in the plans (good status of water bodies and good potential status of artificial or heavily modified bodies). The results achieved after the previous planning cycles show a clear failure to meet the quality objectives, for both surface water bodies and groundwater and the remaining bodies (coastal and transitional waters). It will be necessary to make a major investment effort in wastewater treatment and discharge control in order to achieve the environmental objectives and avoid the investigation and sanction procedures initiated by the EU and other future ones that will foreseeably be initiated.
- The revision of Directive 91/271/EEC on urban wastewater treatment will require special attention and additional investment in this area, establishing new obligations for efficiency and energy neutrality in Waste – Water Treatment Plants (WWTPs) and sewage networks, reuse of wastewater, control of overflows in collectors and storm tanks, etc., and more rigorous control and monitoring.
- Groundwater presents specific problems, mainly reflected in situations of overexploitation (non-compliance with good quantitative status) and, in particular, diffuse pollution caused by certain agricultural and livestock farming activities. A large number of groundwater bodies have been declared vulnerable, reaching nitrate concentration levels significantly higher than those permitted. The establishment and control of codes of good agricultural practices by the corresponding responsible Administrations and their coordination with the Hydrographic Confederations and basin bodies of the River Basin Districts (Demarcaciones Hidrográficas) of intra-community basins is essential.

Governance

- The pioneering creation of the Confederaciones Hidrográficas (from 1927), with their system of user participation and economic-financial regime of passing on costs to the beneficiaries of State investments, has contributed to a better governance of water management, but requires an appropriate organisational and regulatory update to streamline and make their activity more efficient and flexible, providing them with appropriate technical, human and economic resources, which are clearly insufficient at the present time.
- It is advisable to review the economic and financial regime applicable to the use of water (through tariffs and charges to the beneficiaries of the works that make it possible) and other assets in the public water domain, in order to improve the passing on of the costs of water services to users and the economic resources for financing new investments.
- The State Companies (SEIASA, ACUAES, ACUAMED) play an important role - complementary to the Confederaciones Hidrográficas - in leveraging public resources and increasing management capacity in the incorporation of European funds, contributions from users and private financiers.

Reasons to invest in water and its profitability

- Spain's level of investment in this field is clearly lower than that of other European countries, whose productive sectors compete with those of our country.

- The profits and balance of payments performance of sectors such as agriculture (food exports), tourism and the service sector in general are highly dependent on water as an essential input for their development. Investment shortfalls in water will have a negative impact on these sectors.
Focus on the food - water - energy nexus
- As has been reiterated, due to the irregularity of its hydrological regime, Spain has needed a greater number of hydraulic infrastructures than most European countries in order to reach similar levels of water security, so that it would be possible to develop its productive sectors and meet the needs of the population. A large part of this heritage especially that of dams and reservoirs is at risk due to deficient investment in their maintenance (estimated multiplier effect of over 20).
- Boosting the process of digitalisation of the administrative procedures of the Hydrographic Confederations and other institutions of the Public Water Administration and the incorporation of modern technologies in the operation of hydraulic infrastructures (Automatic Hydrological Information System AHIS), reuse and desalination.
- The role of public-private participation should be strengthened in those actions with high economic and financial profitability, and public action should be strengthened in investments of marked social interest and of more uncertain economic profitability.

For all these reasons, in a country so marked by water irregularity and so scourged by extreme phenomena, water must become a priority on Spain's political and social agenda. Water is a matter of state and must be a key issue in the next decade. We can no longer live on the rents of the inheritance received. Plans must really be turned into actions and a large part of the welfare society that we have worked so hard to achieve is at stake.

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Statement on the use of Artificial Intelligence

Not applicable.

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